



AI-Enhanced Damage Assessment from Satellite Imagery in the Russia–Ukraine Conflict

By Adamo Bovolenta
UP2069775

MSc Data Analytics
Project Unit - M32616 - 2024/25

Supervisor – Dr. Alice Good

Masters Project – September 2025

Declarations

Access Rights (optional)	
	I consent to my dissertation in this attributed format (not anonymous), subject to final approval by the Board of Examiners, being made available electronically in the Library Dissertation Repository and/or Department/School/Subject Group digital repositories. Dissertations will normally be kept for a maximum of ten years.
	I understand that if I consent, this dissertation will be accessible only to staff and students for reference only.
	This permission may be revoked at any time by e-mailing data-protection@port.ac.uk .
Ethics	
	I understand that ethical approval will be needed for this project regardless of whether I carry out primary research or not.
	I understand that not obtaining ethical approval is grounds for academic misconduct and may result in my failing my project module.
	I have received ethical approval for this work and include my ethics certificate at appendix A.
	If I conducted primary research, I have included all raw data including research documentation (e.g. survey and participant sheet). I understand that not including the raw data can mean potentially failing the project as this is a mandatory requirement.
	If I have carried out primary research involving employees of a company, I have included evidence i.e. a letter or email, showing that I have the company's permission to involve their staff in my research.
Authenticity of work	
	I confirm that I have read the University rules in respect of plagiarism and student misconduct.
	I declare that this work is substantially my own. Any material from an external source (e.g. text, images and source code) is appropriately cited and referenced.
	For engineering projects, I am aware that web-based, website building tools and other content management systems to build artifacts cannot be used. I have included original source code that I wrote myself within the submission.
Use of AI	

Artificial Intelligence (AI), such as ChatGPT, can be used as a tool to help assist and inform the initial development of your work but it is not a replacement for your own critical thinking and analysis. We require that your work is your own original content, demonstrating your knowledge, skills, and critical thinking abilities. To that end, AI-generated content must not be included in any work you submit for assessment unless suitably referenced. Not doing so is a breach of the University's Academic Regulations as outlined in the [Student Conduct Policy](#) and constitutes an assessment offence.



I have read the [University AI policy](#) and if relevant, acknowledged the use of AI in my work using the [appropriate referencing](#).



I understand that not appropriately referencing the use of AI is an assessment offence equal to plagiarism and may result in a penalty which remains on my student record.

If relevant, please provide details of how you have used AI:

I used AI tools solely to improve wording (grammar, clarity, and phrasing), to format and order the reference list, and to obtain debugging suggestions for code (such as syntax fixes, refactoring, addressing indenting issues). The tool was not used to generate substantive content, design methods, perform analyses, create results, write sections of the dissertation, or invent and/or choose references. I reviewed and verified every AI-assisted edit. I remain fully responsible for the accuracy, originality, and integrity of the work.

Name: Adamo Bovolenta

Date: 17/09/2025

Abstract

This dissertation addresses the urgent need for faster, more consistent assessments of building damage in active conflict zones, where manual mapping cannot keep pace with events. To this end, a two-stage approach was designed: first, satellite images are used to outline building footprints; second, each footprint is assigned a damage level (no damage, minor, major or destroyed). The system builds on models trained on a large natural-disaster dataset and is fine-tuned using conflict-specific examples, with adjustments to handle the imbalance between intact and damaged buildings. In controlled tests on Ukrainian data the system performs well, producing clear building outlines and reliably identifying catastrophic damage. However, its accuracy drops when applied to unprocessed scenes from commercial providers. Variations in resolution, sensor pipelines and urban form make the models less consistent, and performance is weakest when distinguishing between minor and major damage. These limitations mean the tool is best used to support, rather than replace, expert judgement. Practical improvements, such as combining radar and optical sensors, incorporating broader context and explicitly handling differences between data sources, could make the method more robust in the field. These results add to the growing evidence that deep learning can meaningfully support conflict-damage assessment, while still requiring human oversight. The work clarifies the gap to robust, real-world deployment while demonstrating considerable capability under controlled conditions.

Keywords: Building Damage Assessment; Change Detection; Semantic Segmentation; Siamese U-Net; Sentinel-1; Sentinel-2; xBD; Russia-Ukraine Conflict; Remote Sensing; Transfer Learning; Domain Adaptation; Class Imbalance; Deep Learning; Humanitarian Mapping.

Acknowledgements

To express my gratitude in this dissertation, I would like to give special thanks to my father, whose resolute support and encouragement have sustained me throughout this project. I am equally grateful to my mother for always checking on me during the difficult moments and offering reassurance when it was most needed. I also want to thank my partner, who, as a fellow master's student, continually motivated me and challenged my critical thinking. Their combined support and understanding have been invaluable to the completion of this project.

Table of Contents

1	INTRODUCTION	1
1.1	BACKGROUND	1
1.2	MOTIVATION FOR THE RESEARCH	1
1.3	RESEARCH QUESTIONS AND OBJECTIVES	2
	Objective 1: Model Development and Architecture Comparison	2
	Objective 2: Transfer Learning Protocol Validation	2
	Objective 3: Comprehensive Performance Evaluation.....	3
	Objective 4: Reproducible Pipeline Implementation.....	3
	Objective 5: Methodological Documentation and Best Practices	3
1.4	DISSERTATION STRUCTURE	3
2	LITERATURE REVIEW	4
2.1	THE RESEARCH PROCESS.....	4
2.2	MANUAL VISUAL ANALYSIS AND EXPERT INTERPRETATION	4
2.3	EARLY AUTOMATED APPROACHES: FROM RULE-BASED DETECTION TO ADVANCED CHANGE ANALYSIS.....	5
2.4	HYBRID AUTOMATED-MANUAL CROWDSOURCED MAPPING.....	6
2.5	THE DEEP-LEARNING REVOLUTION: FROM SYRIAN CONFLICT TO MODERN AI ARCHITECTURES ..	7
2.6	MULTI-SOURCE FUSION AND SAR INTEGRATION IN URBAN CONFLICT MONITORING.....	9
2.7	FROM UKRAINE TO FOUNDATION MODELS: MODERN AI DEPLOYMENT IN ACTIVE CONFLICTS.....	10
2.8	MODERN SAR-BASED AUTOMATED CHANGE DETECTION: UNSUPERVISED METHODS TO OPERATIONAL TRADE-OFFS	10
2.9	CURRENT STATE AND FUTURE DIRECTIONS	11
3	METHODOLOGY	13
3.1	INTRODUCTION	13
3.2	METHODOLOGICAL DECISIONS.....	14
3.3	RESEARCH HYPOTHESES	14
3.4	METHOD LIMITATIONS AND ETHICAL IMPLICATIONS	15
3.5	DATASET ANALYSIS	16
3.6	PREPROCESSING, HARMONISATION, AND AUGMENTATION	18
	3.6.1 Data Handling for Segmentation.....	18
	3.6.2 Data Handling for Classification.....	18
3.7	EXPERIMENTAL DESIGN	20
	3.7.1 The segmentation step: Pretraining on xBD	20
	3.7.2 The Segmentation Step: fine-tuning on damage_assessment_ukraine	22
	3.7.3 The Classification Step: pretraining on xBD	22
	3.7.4 The classification step: Fine-tuning on damage_assessment_ukraine	27
4	EXPERIMENTAL RESULTS AND DISCUSSION	28
4.1	PERFORMANCE OF THE PROPOSED TWO-STAGE PIPELINE.....	28
	4.1.1 Segmentation Stage: ResUNet Performance in Pre-training and Fine-Tuning.....	28
	4.1.2 The Classification Stage: Pretraining and Fine-Tuning	35
4.2	ANALYSIS AND DISCUSSION	45
	4.2.1 Addressing the Research Questions	45
	4.2.2 Theoretical and Practical Implications.....	46
	4.2.3 Limitations of the Research	47
5	CONCLUSION	48
6	REFERENCES	50
7	APPENDIX A.....	60
8	APPENDIX B.....	61

Table of Figures

FIGURE 1 - RESUNET GENERAL ARCHITECTURE (SOURCE: N.TOMAR, 2021)	20
FIGURE 2 - EARLY FUSION NET ARCHITECTURE (SOURCE: BOULAHIA ET. AL, 2021)	23
FIGURE 3 - SIAMESE U-NET ARCHITECTURE (SOURCE: SONG ET. AL, 2023)	24
FIGURE 4 - LOSS TREND DURING EPOCHS, SIAMESE U-NET TRAINING PHASE	29
FIGURE 5 - VALIDATION IoU AND DICE @ 0.5 DURING EPOCHS	29
FIGURE 6 - THRESHOLD SWEEP (PRETRAINING) FOR MASK BINARIZATION: MEAN IoU AS A FUNCTION OF PROBABILITY THRESHOLD T	30
FIGURE 7 - OPTICAL IMAGES, RESPECTIVE GROUND TRUTH MASKS, AND PREDICTED MASKS @ T=0.90	31
FIGURE 8 - OVERLAYS OF GROUND TRUTH BORDERS (RED) AND PREDICTED BORDERS (GREEN)	31
FIGURE 9 - VALIDATION IoU AND DICE, @0.5, FINE-TUNING ON DAMAGE_ASSESSMENT_UKRAINE	32
FIGURE 10 - LOSS TREND DURING EPOCHS, SIAMESE U-NET FINE-TUNING PHASE	32
FIGURE 11 - THRESHOLD SWEEP (FINE-TUNING) FOR MASK BINARIZATION: MEAN IoU AS A FUNCTION OF PROBABILITY THRESHOLD T	33
FIGURE 12 - QUALITATIVE EVALUATION OF FINE-TUNING SEGMENTATION: OPTICAL IMAGE, GROUND-TRUTH MASK, AND PREDICTED MASK @T=0.90	34
FIGURE 13 - OVERLAYS OF GROUND TRUTH BORDERS (RED) AND PREDICTED BORDERS (GREEN), FINE-TUNING STAGE	35
FIGURE 14 - TRAINING AND VALIDATION: ACCURACY AND LOSS DURING EPOCHS	38
FIGURE 15 - PRETRAINING MACRO-F1 DURING VALIDATION	39
FIGURE 16 - TRAINING AND VALIDATION: ACCURACY AND LOSS DURING FINE-TUNING	40
FIGURE 17 - VALIDATION MACRO-F1 DURING EPOCHS, FINE-TUNING STAGE	41
FIGURE 18 - VISUALIZATION OF THE INFLUENCE OF BIAS	43
FIGURE 19 - QUALITATIVE EXAMPLES OF FOOTPRINT SEGMENTATION AND BUILDING-WISE DAMAGE CLASSIFICATION ON UNPROCESSED PRE/POST SATELLITE IMAGERY	44

Table of Tables

TABLE 1 - ARCHITECTURAL CHOICES FOR THE CLASSIFICATION STAGE IN THE TWO-STAGE PIPELINE	24
TABLE 2 - BEST VALIDATION METRICS DURING PRETRAINING (SEGMENTATION ON xBD)	30
TABLE 3 - EVALUATION METRICS ON THE HELD-OUT TEST SET, DAMAGE_ASSESSMENT_UKRAINE FINE-TUNING	33
TABLE 4 - COMPARISON OF EARLY-FUSION & SIAMESE U-NET EVALUATION METRICS	36
TABLE 5 - COMPARISON OF PER-CLASS METRICS	37
TABLE 6 - EARLY-FUSION CONFUSION MATRIX	38
TABLE 7 - SIAMESE U-NET CONFUSION MATRIX	38
TABLE 8 - EVALUATION METRICS OF THE HELD-OUT TEST SET	39
TABLE 9 - PER-CLASS EVALUATION METRIC F THE HELD-OUT TEST SET	40
TABLE 10 - PER-CLASS EVALUATION METRICS OF THE TEST SET+TTA	41
TABLE 11 - CONFUSION MATRIX OF THE EVALUATION OF TEST SET+TTA	42
TABLE 12 -EVALUATION METRICS OF THE TEST SET+TTA+BIAS	42
TABLE 13 - CONFUSION MATRIX OF THE EVALUATION OF TEST SET+TTA+BIAS	43

1 Introduction

1.1 Background

During armed conflicts, commanders and civil-protection agencies require up-to-date intelligence on strike effectiveness, potential civilian or defender casualties, and priority areas for immediate intervention (*International Committee of the Red Cross [ICRC], 2024*). Traditional workflows such as field reports, analyst-driven photo interpretation, and ad-hoc surveys, are time-consuming and unsafe in contested zones. Conflict is a driver of social and political change and therefore demands proactive, structured approaches to management (*Fisher et al., 2000*). Delays in damage assessment can cost lives and impede the efficient allocation of scarce resources (*ICRC, 2024; Fisher et al., 2000*).

1.2 Motivation for the Research

This research focuses on automated damage assessment in Ukraine, due to its ongoing conflict with Russia, to respond to a critical gap in conflict intelligence: the need for rapid, accurate, and scalable evaluation of strike impacts to inform both defensive and humanitarian decision making. Advances in deep learning, particularly in semantic segmentation and damage-classification of high-resolution satellite and Unmanned Aerial Vehicle (UAV) imagery, offer a pathway to automate and accelerate this workflow (*Zhao & Morikawa, 2024*). By training convolutional neural networks to delineate building footprints and then classify their level of damage (e.g. undamaged, minor, major, destroyed), automated systems can produce quantitative damage maps within minutes of image acquisition (*International Committee of the Red Cross [ICRC], 2024*). Such maps not only verify the success of defensive strikes but also flag zones where civilian harm is most likely, enabling rescue teams to prioritize life-saving interventions and planners to sequence infrastructure repair efficiently (*Fisher, 2000*). As outlined by Fisher, embedding these technical advances with conflict-management frameworks further ensures that assessment accounts for the human dynamics of conflict and pragmatic action strategies.

However, deep learning for war-damage assessment faces challenges that span data availability, imaging conditions, model design, operational constraints, and ethical considerations (*Stewart & Hinds, 2023*). Conflict environments often lack large, high-quality labelled datasets and present extreme domain shifts, such as occlusion by smoke, debris, and rapidly evolving front lines, that standard segmentation classification networks struggle to generalize to. Multimodal sensor fusion (Optical, Synthetic-Aperture Radar) introduces registration and noise issues, while computational limits and limited ground truth verification obstruct real-time deployment at scale (*Braik & Koliou, 2024*). Additionally, adversarial conditions and ethical stakes (misclassifying damage can misdirect relief or military actions) demand robust uncertainty estimation and human-in-the-loop validation. Operational guidance and risk framing from humanitarian practice also inform this study, including International Humanitarian Law (IHL) considerations and cautions around algorithmic decision-making in conflict settings (*International Committee of the Red Cross, 2024; Stewart & Hinds, 2023; Center for Strategic and International Studies, 2024; Avtar et al., 2021*). This study investigates whether a two-stage deep-learning pipeline can produce decision-ready damage maps from satellite imagery in the context of the Russia-Ukraine war.

1.3 Research Questions and Objectives

This study therefore asks four research questions:

RQ1 - Can a Deep Learning system reliably find buildings in fresh/unseen conflict imagery?

RQ2 - Can it grade each building's damage at a level that would support real decisions?

RQ3 - Which modelling and data handling choices most improve reliability across new places and conditions?

RQ4 - Is the two-stage pipeline sufficiently robust for real-world use, and what human checks are needed?

This study seeks to answer the above questions by achieving the following aim and objectives.

The aim of this study is to construct and validate an end-to-end, two-stage pipeline that first segments building footprints in satellite imagery and then assigns damage levels following the standard xView2/xBD four-class taxonomy (undamaged, minor damage, major damage, destroyed). The target performance thresholds are ≥ 0.65 mean IoU (Intersection-Over-Union) for segmentation and ≥ 0.70 F1 for damage classification.

Objectives

Objective 1: Model Development and Architecture Comparison

Develop and empirically compare two architectural approaches for conflict damage assessment:

- A U-Net/ResUNet with pretrained encoders for building segmentation
- Two damage classification variants: early-fusion (6-channel concatenated) and Siamese architectures
- Evaluate the accuracy-efficiency trade-off through validation macro-F1 and inference throughput metrics

Objective 2: Transfer Learning Protocol Validation

Implement and assess a two-stage transfer learning strategy:

- Stage A: Pretrain and compare Early-Fusion Network and Siamese U-Net damage classifiers on the xBD natural disaster dataset
- Stage B: Fine-tune the top performing model on Ukraine conflict data using differential learning rates
- Quantify the domain adaptation benefit through ablation studies comparing transfer learning versus Ukraine-only baselines

Objective 3: Comprehensive Performance Evaluation

Establish robust evaluation protocols aligned with humanitarian applications:

- Segmentation metrics: mean IoU, precision, and recall on building delineation
- Classification metrics: macro-F1, per-class precision/recall, and confusion matrices for four damage levels (undamaged, minor, major, destroyed)
- Generalization assessment: held-out scene testing with qualitative failure analysis

Objective 4: Reproducible Pipeline Implementation

Create an end-to-end automated pipeline ensuring scientific reproducibility:

- Implement leak-proof data splits at scene/geographic level
- Document complete training protocols including hyperparameters, seed control, and early stopping criteria

Objective 5: Methodological Documentation and Best Practices

Produce comprehensive documentation meeting academic and operational standards:

- Complete methodology with ablation studies examining architectural choices, loss functions, and post-processing effects
- Address ethical considerations including data attribution, uncertainty quantification, and human-in-the-loop validation requirements
- Provide clear model selection protocols using validation-only tuning with untouched test sets

1.4 Dissertation Structure

The remainder of this dissertation is structured as follows. Chapter 2 reviews prior work on wartime damage assessment, tracing the evolution from manual visual analysis and early automated approaches through hybrid crowdsourced mapping, the deep-learning revolution in conflict monitoring, multi-source fusion with SAR integration, and modern AI deployment from Ukraine to foundation models, concluding with the current state and future directions. Chapter 3 presents the methodology, including research hypotheses, dataset analysis, preprocessing and augmentation strategies, and the complete experimental design for the two-stage pipeline proposed. Chapter 4 reports experimental results and discussion, detailing the performance of both segmentation and classification stages, addressing the research questions, and examining theoretical implications and limitations. Chapter 5 concludes with principal findings and directions for future work. References are provided in Chapter 6, followed by Appendices A and B containing respectively the Certificate of Ethics Review and the Project Specification Document.

2 Literature Review

2.1 The research process

To make the review reproducible, the search strategy relied on major academic databases such as IEEE Xplore, Scopus, Web of Science, arXiv, ResearchGate, Google Scholar, ScienceDirect, utilizing keyword combinations including "building damage," "war," "conflict," "AI," "deep learning," "segmentation," "classification," and specific conflict identifiers (Syria, Yemen, Ukraine, Myanmar, Israel-Gaza). The search was conducted during May-June 2025.

The inclusion criteria prioritized peer-reviewed publications demonstrating explicit use of optical, radar, or multispectral imagery for war-related damage assessment with clear model architectures and evaluation methodologies. Studies were excluded if they offered purely qualitative assessments, lacked clear task definitions, or focused exclusively on natural disasters without transfer learning relevance. Grey literature from UN agencies and NGOs (e.g., UNOSAT) captured operational methods absent from journals.

Data extraction captured conflict context, sensor specifications, training data characteristics, model architectures, loss functions, evaluation metrics, and performance outcomes. Quality assessment evaluated cross-validation protocols, class imbalance handling, and comprehensive metric reporting. The synthesis organized findings by pipeline architecture, geographic region, and methodological innovations (data augmentation, boundary losses, transfer learning). This literature review informed the research hypotheses and guided methodological decisions regarding architecture selection, loss function design, and evaluation metrics throughout the experimental framework.

2.2 Manual Visual Analysis and Expert Interpretation

Before automated systems, conflict damage mapping largely depended on manual visual analysis by intelligence analysts and remote-sensing specialists. During late-1990s and 2000s campaigns (e.g., Kargil 1999; Iraq 2003), analysts compared pre- and post-event satellite scenes to delineate collapsed buildings, crater fields, and infrastructure damage without algorithmic support (*United Nations Institute for Training and Research [UNITAR], n.d.; European Commission Joint Research Centre [JRC] et al., 2010; Kerle, 2011; UNOSAT, 2006*). The workflow was judgment-driven: operators traced building outlines, interrogated multi-temporal cues, and assigned damage categories by consensus. Organizations such as the United Nations Operational Satellite Applications Programme (UNOSAT) produced detailed, analyst-curated damage inventories, most notably after the 2006 Lebanon War, through systematic image interpretation (*UNOSAT, 2006*). These authoritative layers later became valuable training and validation resources for automated approaches and benchmarking studies (*Kerle, 2011*). Efforts in collaborative web mapping sought to scale expert practice while preserving human oversight, prototyping interfaces and task flows that anticipated later “human-on-the-loop” designs (*Fechner et al., 2015; Knoth et al., 2018; Humanitarian OpenStreetMap Team, 2025*). Manual analysis, however, faces well-known constraints. Throughput is limited, updates are slow under surge demand, and judgments can vary across analysts and under time pressure. Inter-annotator disagreement in damage labels, especially at class boundaries, has

been documented, complicating reliability and downstream evaluation (*Manzini et al., 2025*). Small co-registration errors between pre/post images can also be mistaken for real change, inflating false positives or fragmenting objects (*Sundaresan et al., 2007*). Even with mature coordination practices in humanitarian mapping, capacity limits and quality-assurance expenses restrict city- or country-scale coverage during protracted crises. As conflicts lengthen and imagery volumes rise, the latency between acquisition and analyst-ready products becomes a critical operational bottleneck for emergency management and needs assessments (*UNITAR, 2014; European Commission JRC et al., 2010*).

Between pure photo-interpretation and modern deep learning, Object-Based Image Analysis (OBIA) emerged as a bridge technology. Baatz and Schäpe's multiresolution segmentation, popularized through Definiens/eCognition, introduced scale-controlled "objects" as the fundamental mapping unit (*Baatz & Schäpe, 2000*). The approach matured into a multi-resolution, object-oriented paradigm (*Benz et al., 2004*), was formalized under the GEOBIA label (*Hay & Castilla, 2008*), and synthesized in Blaschke's reviews that traced both uptake and limitations in remote sensing (*Blaschke, 2010, 2013*). By grouping pixels into coherent objects before classification, OBIA reduces the "salt-and-pepper" noise of per-pixel maps and supports rule sets that mirror analyst heuristics, combining spectral cues (e.g., roof tone) with spatial context (shape, compactness, adjacency to shadows or roads), while integrating cleanly with vector layers such as building footprints and parcels. OBIA's limitations are equally clear. Results are sensitive to segmentation scale; the transferability of rule sets across cities or sensors is uneven; and small pre/post misregistration can manifest as spurious change along object boundaries. Operational assessments managed these risks through confidence reporting, cross-source verification, and consensus protocols when aggregating multiple analysts' labels (*UNITAR, n.d.; European Commission JRC et al., 2010*). In practice, OBIA provided a pragmatic intermediate step, more systematic and scalable than fully manual labelling yet still transparent and rule-driven, setting the stage for automated change detection and, later, deep learning. This progression from manual interpretation to object-based analysis established both the methodological scaffolding and many of the validation datasets on which contemporary, AI-driven conflict damage assessment now builds.

2.3 Early Automated Approaches: From Rule-Based Detection to Advanced Change Analysis

By the mid-2000s, analysts began to augment manual interpretation with rule-based automation, laying the plumbing for today's damage-mapping pipelines. During the Darfur conflict (Sudan, ~2003–2008), researchers and human-rights organizations used satellite time series to flag village destruction with simple spectral logic: large drops in vegetation or brightness, the appearance of burn scars, and hot-spot anomalies in thermal channels (*Ashraf, Brabyn, & Hicks, 2013; Marx & Loboda, 2013*). A common workflow emerged. First, coarse screens on moderate-resolution imagery (e.g., Landsat 30 m) identified candidate areas via differenced indices; these were cross-referenced against active-fire detections to isolate likely events (*Giglio, Schroeder, & Justice, 2016*). Second, analysts performed targeted visual checks in very-high-resolution (VHR) imagery to confirm destroyed structures and map extents (*Marx & Loboda, 2013; Lin, 2024*). Variants of this "algorithmic triage plus human confirmation" pattern were formalized as early-warning systems that operationalized index thresholds and scheduled spot checks, anticipating later crowd-assisted and supervised-learning frameworks such as the technique adopted during the 2017 Rohingya crisis in Myanmar's Rakhine State,

where MODIS/VIIRS thermal anomalies guided analysts to candidate burn sites that were then verified with VHR optical data (*Giglio et al., 2016; Yale Humanitarian Research Lab, n.d; Ashraf et al., 2013*). These rule-based methods proved pragmatic when ground access was impossible and timeliness mattered. Their limits were equally clear: moderate spatial resolution curtailed building-level specificity; urban clutter, partial roof damage, shadows, and debris camouflage confounded simple thresholds; and outputs typically highlighted affected zones rather than reliable per-building labels.

To limit false positives and extract more structure from multiband stacks, early studies moved beyond raw image differencing to multivariate change detectors. Change Vector Analysis (CVA) summarized both magnitude and direction of spectral change in feature space (*Malila, 1980*), while Multivariate Alteration Detection (MAD) used canonical correlations to emphasize change and suppress stable background (*Nielsen, Conradsen, & Simpson, 1998*). For fire-related destruction, NBR/dNBR became standard for burn-severity mapping from NIR/SWIR bands and their temporal differences (*Delcourt et al., 2021*). Thresholds shifted from ad-hoc cut-offs to statistical rules, notably Otsu's histogram method and CUSUM change-point tests, yielding more reproducible alerts (*Otsu, 1979; Page, 1954*). Recent CUSUM adaptations underscore continued value for operational change alerts (*Sabir et al., 2024; Shen et al. 2024*). Analysts then applied morphological filters to de-speckle outputs and, where building footprints existed, aggregated pixel responses to objects to reduce salt-and-pepper noise while balancing sensitivity and precision (*Baatz & Schäpe, 2000; Benz, Hofmann, Willhauck, Lingenfelder, & Heynen, 2004; Hay & Castilla, 2008; Blaschke, 2010*). These methodological steps mattered operationally. The fusion of index-based screening, multivariate statistics, and object-level smoothing improved signal-to-noise under real-world conditions and founded an operational pattern that still persists: algorithmic triage to window vast search areas, followed by human verification and curation. The approach also institutionalized data practices, maintaining time-stamped stacks, tracking confidence, and documenting verification, that carried into later crisis-mapping efforts and UN-led assessments (*European Commission Joint Research Centre [JRC] et al., 2010*). Finally, the coexistence of automated alerts with expert QA fostered the collaborative tooling explored in web-mapping applications, where human oversight remained central even as automation advanced (*Knoth, Slimani, Appel, & Pebesma, 2018*). These methodological advances were crucial because they materially affected damage map quality. The evolution from simple thresholding to statistically grounded, object-integrated approaches established operational frameworks combining multiple data sources, statistical optimization, and human verification (*Knoth, Slimani, Appel, & Pebesma, 2018*), principles that would guide the transition to machine learning methods.

2.4 Hybrid Automated-Manual Crowdsourced Mapping

As the limits of purely algorithmic screening became clear, researchers coupled machine efficiency with human judgment. A landmark example is the Darfur web-mapping study by Knoth, Slimani, Appel, and Pebesma (2018), which first prioritized image tiles automatically and then routed them to volunteers for review. In a controlled experiment, algorithmic triage increased the number of detected target objects by 70.7% compared with unguided mapping, direct evidence that machine-in-the-loop tasking can substantially raise human productivity (*Knoth et al., 2018*). Parallel HCI prototypes demonstrated how lightweight web tools could structure tasks, surface uncertain cases, and capture provenance, anticipating later analyst-in-the-loop platforms (*Fechner, Wilhelm, & Kray, 2015*).

In disaster and conflict contexts, micro-tasking models matured quickly after the 2010 Haiti earthquake. GEO-CAN and allied initiatives operationalized crowdsourced image interpretation to grade building damage at scale, practices documented in post-disaster needs assessments and methodological commentaries (*European Commission Joint Research Centre [JRC] et al., 2010; Kerle, 2011*). These workflows established now-standard elements: splitting imagery into small cells; asking volunteers for simple binary or ordinal judgments; collecting multiple, independent votes per tile; and filtering noise via consensus schemes and expert adjudication. Civil society groups later amplified this pattern. For example, Amnesty International’s Decode Darfur mobilized tens of thousands of volunteers to compare multi-temporal satellite chips and flag destroyed settlements, with built-in tutorials and quality controls to mitigate common errors (*Amnesty International, 2016*). Humanitarian OpenStreetMap Team (HOT) similarly scaled volunteer coordination and QA around tasking managers, demonstrating how open communities could supply rapid, structured annotations for crisis response (*Humanitarian OpenStreetMap Team, 2025*). A central design challenge was where to spend limited human attention. Hu, Janowicz, and Couclelis (2017) proposed an information-value strategy to order tasks, steering volunteers toward tiles where their judgments would most improve the map. This “value-of-information” logic, to maximize improvement per unit of human effort, supports contemporary auto-ranking and active-sampling in interactive mapping systems, ensuring that humans validate the most uncertain or consequential cases first (*Hu et al., 2017*). The scheduling approach is the same whether the triage signal is derived from trained models, simple heuristics (like index thresholds), or information (such as recency or proximity to known events).

Hybrid pipelines also imported governance lessons from humanitarian mapping: publish confidence scores, track provenance of each decision, and expose uncertainty explicitly to downstream users. Post-Haiti operational reports emphasized checklists for analyst training, double-entry for sensitive sites, and escalation paths for disagreements, process features that were later embedded in volunteer platforms and research prototypes alike (*European Commission JRC et al., 2010; Kerle, 2011*). The result was a pragmatic division of labour: machines expanded the search space and ranked candidates; humans provided verification, disambiguation, and contextual judgment, particularly for borderline damage classes, cloud-shadow confusion, or registration artefacts. By 2017–2018, these ingredients merged into human-in-the-loop systems capable of operating with limited ground access, addressing the scarcity of verified labels, and responding at the cadence required in fast-moving crises (*Knoth et al., 2018; Humanitarian OpenStreetMap Team, 2025*). Crucially, they laid much of the groundwork for today’s supervised and active-learning pipelines: consensus-labelled tiles seeded training sets; tasking managers became data-collection front ends; and triage scores evolved into model-driven uncertainty estimates that decide which samples to annotate next. In short, the hybrid era translated crowdsourcing from ad-hoc volunteerism into repeatable, auditable annotation pipelines, preserving human agency while making large-scale damage assessment feasible under real operational constraints (*European Commission JRC et al., 2010; Kerle, 2011; Fechner et al., 2015; Knoth et al., 2018*).

2.5 The Deep-Learning Revolution: From Syrian Conflict to Modern AI Architectures

The Syrian Civil War catalysed the move from threshold-based screening to modern AI for damage assessment. As imagery volumes and heterogeneity grew after 2011, manual and rule-

based methods reached practical limits, and deep learning gained traction mid-decade. A widely cited milestone is Mueller et al. (2021), who introduced an automated approach for high-resolution imagery that coupled a convolutional network with spatial-temporal post-processing. The first stage estimated per-location probabilities of destruction from multi-temporal inputs labelled with UNOSAT annotations; the second imposed temporal consistency (e.g., “once destroyed, stays destroyed”) and spatial smoothing. This two-stage design tackled severe class imbalance and produced time-resolved maps that closely tracked manual UNOSAT catalogues (Mueller et al., 2021). This shift aligns with broader calls to systematise remote conflict monitoring (Braithwaite et al., 2023).

In parallel, researchers explored non-DL and fusion routes under data scarcity. Ghandour and Jezzini (2018) combined shadow cues with GLCM texture to classify damage in Damascus. Others fused SAR and optical observations with change detection and footprint overlays, often effective locally but harder to generalize across cities, seasons, and sensors (Aimaiti et al., 2022; Zitzlsberger & Podhorányi, 2024; Bui & Le, n.d.).

A step-change in supervision arrived with the xView2/xBD release, which provided a large, labelled corpus of pre/post-disaster imagery and popularised two-stage pipelines: footprint segmentation followed by damage classification (Gupta et al., 2019; Wu et al., 2021). Although oriented to natural disasters, xBD became a pretraining and benchmarking substrate for conflict work; many studies trained on xBD and fine-tuned on conflict-specific data, bridging disaster response and conflict monitoring (Zheng et al., 2024). Architecturally, convergence emerged around encoder-decoder segmentation and Siamese change-detection designs. U-Net variants remain standard for dense labelling, while bitemporal networks specialise the encoder to capture pre/post deltas. Recent options include densely connected Siamese U-Nets and transformer-based change detectors tailored to very high-resolution scenes (Fang et al., 2021; Wang et al., 2023; Song et al. 2023; Qian et al. 2024). Alisjahbana et al. (2024) two-step DeepDamageNet designs echo this theory (Alisjahbana et al., 2024). Since minority categories (e.g., minor vs. major damage) are rare and boundaries are fuzzy, objectives increasingly combine overlap-based and difficulty-aware terms, Dice/Lovász-style components with Focal, Tversky, or hybrid losses, to boost recall on small or ambiguous regions (Abraham & Khan, 2018; Yeung et al., 2022; Yin et al., 2023). Boundary-aware modules and light post-processors further sharpen outlines; dense CRFs remain a practical way to encourage spatial coherence (Krähenbühl & Koltun, 2011; Wu et al., 2021).

Beyond network choice, engineering practice strongly shapes operational performance. Large scenes are tiled into overlapping patches to fit memory, with sliding-window inference and halos to avoid seam artefacts. Augmentation is critical, both at train time and as test-time augmentation (TTA), to stabilise predictions across vendors and seasons (Buslaev et al., 2020; DeVries & Taylor, 2018). Training stability and efficiency benefit from warm restarts, decoupled weight decay, and mixed-precision compute (Loshchilov & Hutter, 2017, 2019; Micikevicius et al., 2018). In many deployed systems, classical image processing still adds value downstream: CRF-based smoothing and topology-aware clean-up provide the crisp delineations needed for reliable cartography (Gupta et al., 2019; Wu et al., 2021). Taken together, these advances shifted the field from hand-tuned rules to validated, scalable AI pipelines capable of processing vast imagery streams with much lower latency. The lesson from the Syrian context generalises: accuracy gains depend as much on careful engineering and validation, tiling, temporal logic, calibration, augmentation, as on headline architecture. Robust pipelines combine supervised learning with principled post-processing and curated labels,

setting the stage for present work on domain adaptation, foundation-model pretraining, and conflict-specific fine-tuning.

2.6 Multi-Source Fusion and SAR Integration in Urban Conflict Monitoring

As urban conflicts intensified and imagery availability grew, damage-mapping increasingly relied on multi-source fusion. Pairing optical imagery with synthetic-aperture radar (SAR) preserves coverage under cloud, smoke, and darkness, conditions common in active theatres. With Sentinel-1 (C-band; 6–12-day effective revisit), rapid SAR alerts can be issued continuously and cross-checked against optical data when available. Fusion typically extends to GIS layers (building footprints, roads, land use) so that pixel-level signals aggregate into building-level evidence and summary statistics that decision-makers can interpret (*Aimaiti et al., 2022; Zitzlsberger & Podhorányi, 2024; Dietrich et al., 2025*). In conflict theatres such as Ukraine and Gaza, coherence loss from SAR, reflecting decorrelation after structural change, has been combined with open building footprints to flag likely collapses, then summarised at object level for reporting (*Dietrich et al., 2025; Scher & Van Den Hoek, 2025a, 2025b; Zitzlsberger & Podhorányi, 2024*). Where high-resolution optical data are scarce or delayed, night-time lights (VIIRS) provide macro-scale proxies for grid failure and population displacement. They are most effective as auxiliary evidence: urban cores saturate, and dimming can reflect curfews, deliberate shutdowns, or generator substitution rather than physical damage (*Chen et al., 2021; Jiang et al., 2017*). Coherence-based “damage proxy maps” have also been evaluated for operational use, underscoring both their promise and the need for careful validation against ground truth and analyst products (*O’Donnell, 2024*). SAR’s resilience comes with processing requirements. Backscatter (σ^0) must be radiometrically calibrated, incidence-angle normalised, terrain-corrected, and de-speckled to stabilise statistics, using filters from adaptive Lee/Frost to more recent non-local means and speckle-whitening approaches (*Chan et al., 2022; Alparone et al., 2024*). Change cues are typically drawn from two families: interferometric coherence, which drops where scatterers decorrelate after structural change, and amplitude/intensity change, which is easier to compute but more sensitive to viewing geometry and environmental variability. Coherence estimation requires windows: small windows are noisy, large windows blur localised damage. In practice, pairwise coherence maps provide early alerts, while short time-series stacks (or a second post-event pass) improve reliability before publishing object-level summaries (*Dietrich et al., 2025; Scher & Van Den Hoek, 2025a, 2025b*).

Fusion improves robustness and interpretability but heightens co-registration risk. Even sub-pixel misalignment between SAR and optical layers can masquerade as change, especially along sharp boundaries (roof edges, narrow streets). Operational pipelines therefore combine precise orthorectification and tie-point refinement with modern cross-modal registration, including learning-based correlation and attention mechanisms, for SAR-optical pairing (*Zhang et al., 2025*). After alignment, auxiliary GIS context (building footprints, transport layers, land use) is routinely used to aggregate pixel-wise evidence into per-building or neighbourhood metrics that better support triage and planning (*Aimaiti et al., 2022; Zitzlsberger & Podhorányi, 2024*).

2.7 From Ukraine to Foundation Models: Modern AI Deployment in Active Conflicts

The war in Ukraine, especially after the February 2022 invasion, has driven some of the broadest operational use of AI-enabled damage mapping to date, simply because the tempo and scale of destruction demanded it (*Dietrich et al., 2025; Zitzlsberger & Podhorányi, 2024; Center for Strategic and International Studies [CSIS], 2024*). Workflows that once lived in research prototypes were pushed into near-real-time pipelines. A useful case study is Risso et al. (2024), who evaluated zero-shot transfer versus conflict-specific fine-tuning on very-high-resolution imagery from Mariupol. Models pretrained onxBD/xView2 could be applied “as is,” but consistent gains came from fine-tuning on local labels, underscoring that conflict damage has signatures, blast patterns, progressive shelling, debris occlusion, not fully captured by natural-disaster corpora (*Risso et al., 2024*).

Related efforts extended the recipe to other theatres. For the October 2023 Israel– Hamas war, Zhao et al. (2024) fused night-time lights (for grid disruption) with VHR optical imagery and a transformer segmenter pretrained onxBD/xView2, then fine-tuned on hand-labelled Gaza samples. In Ukraine, Zitzlsberger and Podhorányi (2024) proposed an ensemble recurrent CNN over Sentinel-1 SAR and Sentinel-2 optical time series; ablations showed optical features drove per-building specificity while SAR maintained coverage through cloud and smoke. Across these studies, a pragmatic pattern emerged: start from a strong disaster-trained encoder, use the sensing mix available in theatre, and fine-tune on a focused set of conflict labels (*Risso et al., 2024; Zitzlsberger & Podhorányi, 2024; Zhao et al., 2024; Wang Y et al. 2025*). Operational deployments also exposed a central pain point: domain shift. Models trained in one region or hazard type often stumble elsewhere because of differences in sensors, vendors, seasons, urban form, and labeling practice. Domain-adaptation methods address this by aligning source and target feature distributions. Moment-matching approaches (e.g., Deep Adaptation Networks with MMD; Long et al., 2015) and CORAL (feature covariance alignment; Sun & Saenko, 2016) reduce statistical discrepancies, while adversarial adaptation trains a domain discriminator, so representations become domain-invariant (Ganin et al., 2016). In damage-mapping specifically, these strategies improve transfer to new events and cities, especially when coupled with limited local fine-tuning (*Wang et al., 2021; Zheng et al., 2024*).

The trajectory from Ukraine to recent Gaza work marks a shift from bespoke, event-specific models toward generalizable encoders with targeted adaptation. The most reliable recipe is clear: pretrain for breadth and speed, fine-tune on a modest local set for fidelity, add explicit domain adaptation when the sensor mix or urban fabric changes, and apply careful calibration so probabilities and thresholds remain trustworthy in operations (*Risso et al., 2024; Zhao et al., 2024; Long et al., 2015; Sun & Saenko, 2016; Ganin et al., 2016*). This blend, pretraining for coverage, adaptation for transfer, and calibration for decision-readiness, defines the present frontier in conflict-zone damage assessment.

2.8 Modern SAR-Based Automated Change Detection: Unsupervised Methods to Operational Trade-offs

Synthetic-aperture radar (SAR) enables rapid, all-weather monitoring of conflict damage through two complementary signal families: interferometric coherence (sensitive to abrupt

structural change) and amplitude/backscatter change. Both support unsupervised and light-supervised workflows. On the unsupervised side, long temporal-arc coherent change detection (LT-CCD) for Gaza builds coherence stacks from Sentinel-1 pairs and flags damage where coherence drops persist across time. Against UN reference data, reported performance includes a $\sim 92.5\%$ true-positive rate and $\sim 1.2\%$ false-positive rate, illustrating that coherence-only methods can deliver high recall with minimal tuning (Scher & Van Den Hoek, 2025a). A second unsupervised option, the Pixel-Wise T-Test (PWTT), applies a two-sample t-test to pre/post amplitude time series to identify statistically significant changes without labels, simple to implement, fast to run, and naturally suited to wide-area screening (Ballinger, 2024a, 2024b).

Light-supervised pipelines trade a small amount of labeled data for building-level products. Dietrich et al. (2025) trained features from amplitude stacks against UNOSAT damage points and aggregated outputs to footprints to yield per-building probabilities at national scale in Ukraine (see also *prs-eth/ukraine-damage-mapping-tool*, *GitHub*, *n.d.*). This pattern, temporal features + compact supervision + footprint aggregation, has become a practical compromise between speed and specificity. Each SAR cue has operational trade-offs. Single-pair coherence can produce Damage Proxy Maps within a day of acquisition, ideal for alerts, but may miss gradual/partial degradation and is sensitive to decorrelation from soil moisture or vegetation. Amplitude change works across differing tracks and longer gaps but is noisier unless radiometrically normalised and de-speckled; multi-temporal stacks help suppress transient noise. Many operations therefore accept a short delay (an extra post-event pass, ~ 6 – 12 days for Sentinel-1) to stabilise stacks before dissemination (Dietrich et al., 2025; Scher & Van Den Hoek, 2025a, 2025b). Turning pixel evidence into actionable building labels requires explicit aggregation rules (e.g., majority vote, or “ $\geq x\%$ of footprint changed”). Thresholds are tuned on validation areas to balance sensitivity and precision, and outcomes depend on footprint accuracy and size: small structures are easy to over-flag; large complexes can show internal change without meeting a “destroyed” criterion. Because these design choices materially affect triage lists, best practice is to publish the time window, pair/stack composition, coherence or amplitude thresholds, building-level decision rules, and simple uncertainty indicators alongside the maps.

Overall, recent SAR work shows complementarity between transparent, training-free detectors (LT-CCD, PWTT) and light-supervised pipelines (Dietrich et al., 2025): the former provide fast, interpretable alerts under cloud and smoke, while the latter deliver stabilised, building-resolved products suitable for planning and ongoing monitoring (Ballinger, 2024a, 2024b; Scher & Van Den Hoek, 2025a, 2025b).

2.9 Current State and Future Directions

War-damage assessment has evolved from expert photo-reading to data-driven pipelines. Most operational systems now follow a two-stage design: first, segment building footprints with U-Net/ResUNet-style encoders–decoders, typically pretrained then fine-tuned, often with overlap/boundary-aware losses and light post-processing to stabilize masks; second, classify damage for each footprint from paired pre/post crops using either early fusion (concatenated features) or Siamese encoders (shared weights). Class imbalance is handled with class-weighted cross-entropy or focal-type losses (e.g., Dice/Focal/Tversky variants) (Abraham & Khan, 2018; Yeung et al., 2022). This workflow uses segmentation metrics such as mean IoU, precision/recall, and for the damage labels emphasizes macro-F1 with per-class precision/recall;

confusion matrices help diagnose “minor vs. major” confusions. Leakage-free geographic splits (scene/city-based) and validation-fixed probability thresholds are used to ensure robust testing (Gupta et al., 2019; Wu et al., 2021).

Several field-tested practices consistently improve performance. Multi-source fusion (optical + SAR, sometimes with night-time lights) provides all-weather coverage that single sensors cannot match (Aimaiti et al., 2022; Zitzlsberger & Podhorányi, 2024; Chen et al., 2021; Jiang et al., 2017). Transfer learning from established corpora such as xBD/xView2 accelerates development, with conflict-specific fine-tuning closing the domain gap (Gupta et al., 2019; Wu et al., 2021). For bi-temporal change, Siamese networks and their recent transformer/attention variants have become standard (Fang et al., 2021; Wang et al., 2023). Temporal modeling, from simple smoothing and “once destroyed, stays destroyed” logic to sequence models, reduces false positives in imbalanced settings (Müller et al., 2021; Zitzlsberger & Podhorányi, 2024). Classical post-processing remains useful: dense CRFs or similar spatial regularisers sharpen boundaries and improve object coherence (Krähenbühl & Koltun, 2011). Training-free SAR baselines such as long-arc coherence change detection and pixel-wise statistical tests provide fast, interpretable alternatives that complement deep models operationally (Ballinger, 2024; Scher & Van Den Hoek, 2025; Dietrich et al., 2025). Foundation-model pretraining and disciplined validation now enable near-real-time, large-area monitoring (Kirillov et al., 2023; Huo et al., 2025).

Despite these advances, building-damage AI is currently built upon natural-disaster datasets such as xBD, which contains 850 736 annotated buildings across 45 362 km² of imagery (Gupta et al., 2019), but this dataset, and most others, focus exclusively on natural disasters. Consequently, models trained on xBD generalise poorly to conflict damage because war-induced destruction has different morphology and evolves continuously rather than discretely. A recent study on conflict zones notes that while very-high-resolution imagery is plentiful for natural disasters, annotated data specific to war scenarios is “comparatively scarce,” deterring the training of deep neural networks for conflict damage (Risso et al., 2022). In fact, xBD remains “the only large-scale dataset” with pre-/post-disaster buildings, and there are no peer-reviewed open datasets with annotated war damage (Risso et al., 2022). This scarcity forces researchers to attempt transfer learning from natural-disaster models, but evidence shows limited success: Risso et al. (2024) Mariupol case study found that state-of-the-art CNNs trained on xBD achieved balanced accuracy of only 59–69 % on conflict. Furthermore, high-resolution satellite data needed for building-level assessment is costly and infrequent, constraining real-time monitoring. Recent foundation models such as IBM and NASA’s Prithvi, pre-trained on 4.2 million samples of Harmonized Landsat-Sentinel-2 data, show promise by improving flood and burn mapping by 15 % (NASA Earthdata, 2024), but these models are still aimed at general Earth-observation tasks and have not been adapted to conflict damage. Together, these gaps highlight the need for new, openly available conflict-specific datasets, tailored architectures (e.g., Siamese or multi-sensor models) and human-in-the-loop workflows to bridge the performance divide between humanitarian and military contexts.

Looking ahead, research is converging on tighter integration between segmentation and damage classification via shared encoders and explicit boundary objectives; longer-horizon temporal models that encode repeated strikes or gradual collapse; and human-in-the-loop systems that prioritize uncertain cases for expert review (Fang et al., 2021; Zitzlsberger & Podhorányi, 2024; Knoth et al., 2018; Hu et al., 2017). Across Syria, Yemen, Sudan, Myanmar, Israel-Palestine, and Ukraine, practice is coalescing around pipelines that fuse multiple data sources, exploit temporal information, and maintain human oversight at key decision points

(Müller et al., 2021; Chen et al., 2021; Jiang et al., 2017; Dietrich et al., 2025). Foundation models and transfer learning reduce spin-up time in new theatres, while SAR-based unsupervised methods keep coverage steady regardless of weather. Evaluation has matured toward object-level and per-class reporting aligned with humanitarian priorities. The remaining work is largely disciplined engineering: careful data curation, leakage-free validation, calibrated and interpretable outputs, and explicit ethical framing so the technology supports, rather than replaces, responsible human judgment (Al Rahrooh et al., 2024). Cross-vendor generalisation remains the rate-limiter; feature-based and discrepancy-based domain adaptation offer practical routes for remote sensing tasks, complemented by adversarial and multi-source strategies (Sun & Saenko, 2016; Long et al., 2015; Ganin et al., 2016; Wang et al., 2021; Martakis et al., 2022).

3 Methodology

3.1 Introduction

This chapter details how the study operationalizes its aim, building and validating a two-stage deep-learning pipeline for conflict-zone damage assessment, and links each research question to a concrete methodological component. The pipeline follows the now-standard pattern in the field: a semantic-segmentation stage to delineate building footprints in unseen satellite imagery, followed by a damage-classification stage assigning labels (undamaged, minor, major, destroyed) to each detected building (Gupta et al., 2019; Wu et al., 2021; Alisjahbana et al., 2024). Segmentation uses encoder–decoder variants proven in related work (e.g., ResUNet/UNet++), while change-aware classification leverages bi-temporal designs, either early fusion or Siamese encoders with shared weights, to increase sensitivity to pre/post differences (Fang et al., 2021; Jha et al., 2019).

The four research questions (whether buildings can be reliably found, whether damage can be graded usefully, which design choices improve reliability, and whether the system is robust enough for operational use) map directly onto the chapter’s sections on data preparation, model design, training protocols, transfer learning, and evaluation. Design choices (architectures, losses, thresholds) are informed by prior literature on damage mapping and imbalanced segmentation, including focal/Tversky/Dice-family objectives used to stabilize minority classes (Abraham & Khan, 2018; Yeung et al., 2022). Evaluation follows common practice: mean IoU (Overlap of predicted mask and ground-truth mask G), precision, and recall for segmentation, and macro-F1 (harmonic mean of precision and recall, with equal weights across the four classes) with per-class precision/recall for damage labels, scene/city-based (leakage-free) splits and thresholds fixed from validation before testing to ensure fair assessment (Gupta et al., 2019; Wu et al., 2021). Transfer learning from xBD/xView2 and conflict-specific fine-tuning are used to manage domain shift, consistent with prior work and domain-adaptation principles (Long et al., 2015; Sun & Saenko, 2016; Ganin et al., 2016; Zheng et al., 2024; Rizzo et al., 2024). Finally, the experimental protocol emphasizes transparency and reproducibility, reporting data provenance, configurations, and metrics in line with emerging best practices for interoperable ML studies (Al Rahrooh et al., 2024). Together, these components provide a clear

basis for answering the research questions and meeting the stated targets (≥ 0.65 mean IoU for segmentation; ≥ 0.70 macro-F1 for multiclass damage classification).

3.2 Methodological Decisions

This project’s approach aligns well with what has been used successfully in Ukraine. In fact, many top solutions for assessing building damage in the literature use the proposed two-stage approach. For example, Deng & Wang (2022) developed a two-stage model called “MBDA-Net” in which stage 1 is an independent U-Net that segments building footprints, and stage 2 is a Siamese U-Net that classifies the damage level using pre- and post-images. By first isolating buildings, the model can focus on each structure’s before/after state to determine if it’s intact, moderately damaged, heavily damaged, or destroyed. This method, trained on xBD, achieved a high localization F1 score of 0.874 and damage classification F1 of 0.754. Two-stage pipelines like this were also reflected in the xView2 challenge winners, many teams first used building segmentation (often with U-Net or Mask R-CNN) and then classification on each building patch, sometimes via a separate network or by adding a classification head to the segmentation model (Gupta et al., 2019; Wu et al., 2021; Fang et al., 2021). Overall, the two-stage approach is considered ideal for building-wise damage assessment. The first stage ensures that only actual building areas are analysed, reducing false positives from background), and the second stage can concentrate on subtle damage features on a per-building basis (Deng & Wang, 2022; Fang et al., 2021; Wu et al., 2021).

In Ukraine, evaluation and validation of models have been a major focus due to the operational importance of results. Common evaluation metrics include precision, recall, and F1-score at the building level. Given the large class imbalance, caused by far more undamaged buildings than damaged, many studies report per-class precision/recall and use a balanced F1 as the main metric (Deng & Wang, 2022; Gupta et al., 2019). Muller et al. (2021) specifically reported precision on positive (destroyed) class in unbalanced samples to give a realistic sense of performance. Intersection-Over-Union (IoU) is used for segmentation quality, often requiring $\text{IoU} > 0.5$ overlap to count a building as correctly localized (Gupta et al., 2019; Wu et al., 2021). The xView2 challenge introduced a composite metric that only counts a detection as correct if the building is correctly localized and the damage class is correct, effectively a strict score. The best models in research and competitions have steadily pushed these metrics higher. For instance, Deng & Wang’s improved U-Net had ~ 0.75 F1 on damage classification, whereas earlier simpler models were in the 0.6 - 0.7 range.

3.3 Research Hypotheses

The study adopts a two-strand quantitative design. Strand one is a systematic review of war-damage assessment research, covering conflicts from early remote-sensing approaches through recent deep-learning models. It identifies common pipelines (segmentation plus per-building classification), data sources, augmentation techniques, and evaluation practices that inform later experimental choices (Gupta et al., 2019; Müller et al., 2021; Zheng et al., 2024). Strand two is an experimental programme using KOlegaBB’s dataset (damage_assessment_ukraine, retrieved from HuggingFace) for segmentation and fine-tuning and the xBD (xView2) dataset for pretraining the classifier. Pre- and post-event images are

harmonised and partitioned into spatially disjoint train, validation and test sets to avoid leakage. The segmentation model, a U-Net variant with a pretrained encoder and optional boundary head, is trained using hybrid loss functions that combine Dice, Focal-Tversky and edge-aware terms. The classifier is evaluated on different backbones and fusion methods for pre/post pairs and trained with cross-entropy or focal losses (*Abraham & Khan, 2018; Yeung et al., 2022; Fang et al., 2021*). Experiments include ablations on data augmentation, loss design and post-processing; comparisons of architectures; and transfer-learning tests. The following hypotheses guide the experiments:

- **Hypothesis 1:** A deep segmentation model with skip connections and boundary-aware loss will achieve an operational mean IoU of ≥ 0.65 on unseen Ukraine scenes. (*Quian et.al 2024; Hu et. al 2024; Appati et. al 2024*)
- **Hypothesis 2:** Pretraining the damage classifier on xBD and fine-tuning on `damage_assessment_ukraine` will yield higher macro-F1, particularly for major and destroyed classes, than training solely on the conflict-specific dataset. (*Gupta et al., 2019; Zheng et al., 2024*).
- **Hypothesis 3:** Hybrid loss functions (e.g. Dice + Focal-Tversky) and realistic augmentation (geometric and photometric) will improve overlap and classification accuracy relative to single-loss or no-augmentation baselines. (*Yeung et. al 2022; Abraham et. al 2018; Yin et. al 2023*)
- **Hypothesis 4:** Post-processing (threshold tuning, small-object filtering) and assembling will provide incremental gains without excessive complexity. (*Gupta et. al 2019; Wang et. al 2025; Ma et. al 2025*)

Threats to validity include class imbalance, domain shift between natural-disaster and conflict imagery, and limited/variable labels; mitigation includes balanced sampling, leakage-free spatial splits, transfer learning, and qualitative overlays for sanity checking (*Müller et al., 2021; Zheng et al., 2024; Al Rahrooh et al., 2024*).

3.4 Method limitations and Ethical Implications

Ukraine-focused datasets pose several limitations. First, imagery is heterogeneous, across sensors, dates, seasons, look angles, and resolutions, which requires normalization and careful co-registration before learning (*Zitzlsberger & Podhorányi, 2024; Aimaity et al., 2022; Zhang et al., 2025*). Second, ground truth is scarce during an active war, so labels come primarily from manual annotation, which can be inconsistent or noisy (*Müller et al., 2021; Gupta et al., 2019; Wang Z et. al 2025*). Third, domain shift is acute: models trained in one city or disaster type often degrade elsewhere due to differences in architecture, materials, or imaging conditions. Mitigations include transfer learning and domain-adaptation methods such as correlation alignment, deep adaptation networks, and adversarial alignment (*Sun & Saenko, 2016; Long et al., 2015; Ganin et al., 2016; Wang et al., 2021; Zheng et al., 2024*). Finally, distinguishing conflict damage from other change (e.g., deliberate demolition for defences, weather-driven deterioration) remains difficult; studies therefore constrain analyses to known conflict periods and/or compare pre-war versus post-war imagery to target conflict-related change specifically (*Müller et al., 2021; Dietrich et al., 2025*).

Ethically and professionally, Ukraine’s case highlights a few points. Consistent with guidance on the use of AI in armed conflict, transparency and harm minimization are essential (*ICRC,*

2023; 2024). This study adopts a conservative release and evaluation protocol: although the `damage_assessment_ukraine` corpus spans three Ukrainian cities, the thesis withholds coordinate metadata, scene identifiers, and city-image mappings; figures are cropped and scaled to explain model behaviour without enabling re-identification. To mitigate bias, especially the risk that disaster-trained models under-count war damage, training and evaluation emphasize per-class reporting and domain adaptation (Gupta et al., 2019; Zheng et al., 2024). Because these maps inform humanitarian triage and reconstruction planning, a human-in-the-loop workflow is retained: models produce preliminary damage layers that are reviewed by analysts, with consensus protocols and confidence reporting adapted from operational mapping practice (Knoth et al., 2018; UNOSAT/JRC/GFDRR/CNIGS, 2010).

3.5 Dataset Analysis

This study draws on two complementary datasets whose roles are deliberately distinct. `damage_assessment_ukraine` provides conflict-domain imagery and labels from the Russia-Ukraine war, which matches the operational context in which the proposed pipeline is intended to function. xBD (xView Building Damage) supplies the scale and label diversity that modern deep models need to learn generalisable damage cues. Used together, they support a two-stage methodology in which models are pretrained on xBD (Gupta et al., 2019) and then fine-tuned and evaluated on `damage_assessment_ukraine`, developed to address the limited availability of publicly labelled data from actual post-disaster scenarios in Ukraine, to test transfer-learning into a genuine conflict setting. The dataset is structured around two complementary tasks essential to this research. The building segmentation component provides pre-event satellite tiles from three strategic locations: Kamianka in Kharkiv oblast, Yakovlivka in Donetsk oblast, and Popasna in Luhansk oblast, each accompanied by binary building masks. The damage assessment component offers individual building crops captured before and after conflict events, complete with footprint masks and four-level damage classifications that align with established xView2/xBD taxonomies, ranging from undamaged through minor and major damage to destroyed (Gupta et al., 2019). The complete dataset includes 169 pre-event tiles designated for segmentation tasks and 2,219 annotated building instances for classification analysis. Geographic distribution varies across locations: Kamianka contributes 62 tiles and 855 buildings spanning from July 2020 to August 2022, Yakovlivka provides 40 tiles and 760 buildings covering September 2019 to August 2022, while Popasna includes 67 tiles and 604 buildings from October 2020 to July 2023. The imagery collection involved extracting post-event scenes from Google Maps and pre-event imagery from Google Earth historical archives using QGIS, with pre-event scenes requiring manual georeferencing. Standardization processes normalized all tiles to 1024×1024-pixel dimensions at approximately 0.33 meter per pixel resolution. Damage labels were assigned through systematic visual inspection following xBD definitional frameworks. Evaluation protocols organize both segmentation and classification subsets into 10-fold configurations, with segmentation folds designed to minimize blank masks while classification folds employ stratification to maintain balanced label distribution across experimental splits.

Several acknowledged limitations characterize `damage_assessment_ukraine`. Pre-event imagery typically exhibits lower quality compared to post-event scenes, creating asymmetrical conditions that models must accommodate. The annotation process involved a limited team without expert adjudication, potentially introducing inconsistencies particularly at boundaries between minor and major damage classifications (see also label-agreement issues noted by

Manzini et al., 2025). Coverage represents purposeful sampling rather than exhaustive documentation, exemplified by partial annotation of Popasna, and not all buildings visible in scenes receive guaranteed labelling. Despite filtering procedures targeting low-visibility cases, residual noise from atmospheric effects, shadows, and occlusion remains possible. Public research access is available for the dataset, though underlying imagery remains subject to Google Maps and Earth usage terms, precluding redistribution or commercial applications. While *damage_assessment_ukraine* provides conflict-domain authenticity, xBD contributes essential breadth and scale. This comprehensive benchmark encompasses pre and post satellite tiles featuring building polygons and unified four-level damage classifications across numerous global natural disasters. The dataset includes approximately 22,000 tiles in 1024×1024 RGB format derived from 19 distinct events, covering roughly 45,000 square kilometres with nearly 850,000 annotated building footprints. Official partitioning follows an 80/10/10 train/validation/holdout split structure. This research utilizes xBD primarily as a pretraining foundation through the xView2 Challenge subset, which maintains the original labelling scheme and diversity while offering practical experimental scale. Dataset availability follows CC BY-NC-SA licensing, requiring attribution and restricting commercial use in reported results (*Gupta et al., 2019*). The input harmonization across sources involves normalizing and resizing all imagery to consistent network dimensions. *Damage_assessment_ukraine* partitions maintain spatial disjunction at scene level to prevent data leakage, with segmentation outputs converted to connected components for extracting precise per-building crops designated for damage classification. The transfer learning approach initializes classification architectures using xBD-trained models, subsequently fine-tuning on *damage_assessment_ukraine* crops to adapt to conflict-specific damage signatures including blast patterns and partial structural failures. This methodology tests whether natural disaster damage features successfully transfer to armed conflict imagery and identify conditions where transfer fails (*Müller et al., 2021; Zitzlsberger & Podhorányi, 2024; Zheng et al., 2024*). Risk management and corresponding mitigation strategies maintain consistency across experimental protocols. Class imbalance, particularly the scarcity of destroyed buildings and occasionally major damage in certain locations, receives attention through stratified folding, balanced sampling, and appropriate loss weighting. Potential label noise and classification ambiguity are addressed by reporting macro-F1 scores with per-class precision and recall metrics, examining confusion matrices, and supplementing quantitative assessments with qualitative map visualizations (*Manzini et al., 2025*). The domain gap between xBD and *damage_assessment_ukraine* receives explicit experimental treatment by comparing Ukraine-only baseline training against xBD-pretrained fine-tuning to identify beneficial or negative transfer effects (*Sun & Saenko, 2016; Long et al., 2015; Ganin et al., 2016; Zheng et al., 2024*). Generalization assessment involves withholding entire scenes and comparing performance across the three *damage_assessment_ukraine* locations, each representing different temporal conditions and destruction patterns (*Zitzlsberger & Podhorányi, 2024*).

3.6 Preprocessing, Harmonisation, and Augmentation

3.6.1 Data Handling for Segmentation

The segmentation pipeline proceeds in two stages: pretraining on xBD to acquire generic building-footprint cues, followed by fine-tuning on `damage_assessment_ukraine` to adapt to conflict-specific imagery. For xBD, polygonal annotations are rasterised into binary masks by prioritising pixel-space WKT when present and otherwise falling back to GeoJSON polygons or image-space bounds; holes in complex structures are supported. Masks encode buildings as 255 and background as 0. A lightweight quality-control pass corrects rare annotation inversions by flipping masks when more than 60% of pixels are non-zero, a conservative choice for urban scenes, where buildings typically occupy far less than half the image; this threshold was selected after inspecting random samples and proved to eliminate evident inversions without harming valid masks. All masks are then normalised to strict binary values, checked to match the paired image size, and saved with a consistent `_mask.png` suffix. Pretraining deliberately uses only pre-disaster images to avoid post-event confounders and to emphasise unobstructed footprints. A scene-level 80/20 split (seed 1337) governs the train/validation partition, with a manifest recording paths and split membership (*Gupta et al., 2019*, for the dataset/taxonomy).

Augmentation (Albumentation 2.0.8) addresses the substantial foreground sparsity inherent to footprint extraction. The pipeline employs `CropNonEmptyMaskIfExists` to bias sampling toward building-containing regions, with a centred crop as a fallback when no valid region is found. Geometric operations are applied jointly to image and mask (with nearest-neighbour interpolation for the latter) and include 90° rotations and horizontal flips at 0.5 probability, vertical flips at 0.3, and a `ShiftScaleRotate` transform constrained to $\pm 8\%$ translation, $\pm 15\%$ scale, and $\pm 15^\circ$ rotation at 0.7 probability. A mild `GridDistortion` ($p=0.2$) introduces additional variability without compromising topological fidelity. Photometric adjustments are image-only and limited in amplitude, brightness and contrast varied within $\pm 15\%$, HSV channel deltas set to 10/15/10, alongside occasional Gaussian blur with 3–7-pixel kernels ($p=0.1$). Inputs are padded before 512×512 crop extraction and are normalised to ImageNet channel statistics (Buslaev et al., 2020; He et al., 2016); masks are then cast to float $\{0,1\}$ for training.

Fine-tuning on KOlegaBB’s dataset preserves geographic separation across cities (Kamianka, Popasna, Yakovlivka) by adhering to the dataset’s fold protocol: folds 2, 3, 5-9 for training, folds 1 and 4 for validation, and fold 0 held out for testing. Because tiles are larger (1024×1024) and contain finer structural evidence, augmentation is more conservative: translations are limited to $\pm 5\%$, scale to $\pm 10\%$, and rotation to $\pm 15^\circ$, with photometric changes reduced to $\pm 10\%$ brightness/contrast and HSV deltas of 8/12/8; blur uses 3-5-pixel kernels at low probability. This balance retains critical edge and texture details while still providing the diversity needed for robust generalisation.

3.6.2 Data Handling for Classification

The classification pipeline operates on paired pre- and post-event imagery and admits scenes only when a complete pair is available together with at least one post-event annotation source. Damage categories are mapped to a canonical four-class scheme {no-damage, minor, major, destroyed} $\rightarrow \{0,1,2,3\}$. Although the loader can rasterise JSON geometry in a fixed priority

order (GeoJSON polygons, then image-space coordinates, then WKT), in practice the `damage_assessment_ukraine` variant predominantly provides single-channel index masks in $\{0,1,2,3,4\}$, where 0 denotes background and 1 - 4 correspond to the four damage levels. Connected components on the positive mask define candidate instances; very small components (<20 pixels) are discarded to suppress annotation noise, corresponding to sub-building fragments below the spatial scale of interest at this dataset’s ground sample distance (the physical mapping is reported in the Appendix). Each retained component is assigned the majority class observed within its extent. Bounding boxes are enlarged by 25% in both dimensions to preserve contextual cues such as debris fields and shadows that assist class disambiguation, a choice that outperformed tighter crops in a small sensitivity check. Identical windows are then extracted from the pre- and post-event images, resized with bilinear interpolation to 256×256 , and saved as RGB PNGs under deterministic names embedding scene core, instance identifier, and class label; a manifest with absolute paths provides reproducible indexing.

Preprocessing for model input mirrors the segmentation normalisation but is tailored to paired change cues. All images are loaded in RGB and normalised with ImageNet statistics to maintain compatibility with ImageNet-pretrained encoders. Geometric transforms are strictly synchronised between pre and post so that change geometry is preserved (a standard practice in Siamese/change-detection pipelines; cf. *Fang et al., 2021*). In training, a shared `RandomResizedCrop` draws windows at scales 0.8-1.0 and aspect ratios 0.75-1.33 before resizing to the input resolution, followed by horizontal and vertical flips with probabilities 0.5 and 0.2 respectively. To simulate occlusions common in post-disaster scenes, a CutOut-style erasure removes one to three rectangular regions (sides 16-38 pixels) with an overall cap of 20% area; by default, the same holes are applied to pre and post to retain alignment, with a small probability (0.2) of unpaired erasures to inject mild asymmetry (*DeVries & Taylor, 2018*). Removed regions are filled with the per-image channel mean to avoid introducing artificial dark artefacts. A light `ColourJitter` (brightness/contrast/saturation 0.1, hue 0.05) is applied independently to each image to model acquisition differences. Validation eschews augmentation beyond deterministic resizing and normalisation, while testing uses a four-way flip-based test-time augmentation (identity, horizontal, vertical, and combined) with logit averaging; validation remains TTA-free for honest model selection. Splitting is performed at the scene level to avoid leakage. For xBD-derived classification crops a 70/15/15 train/validation/test split (seed 42) is used; for `damage_assessment_ukraine`, the fold protocol is preserved, with `fold_1` as validation and `fold_0` as the final test set. Class imbalance is handled primarily at the sampling stage via a tempered weighted sampler with exponent 0.5, which moderates the aggressiveness of inverse-frequency sampling; when effective-number class weights are used at the loss, they are not stacked with the sampler to avoid over-correction (*Cui et al., 2019*). Data loaders employ pinned memory, a fixed batch size (e.g., 64) with `drop_last=True`, deterministic seeds, and conservative worker settings suited to the execution environment.

For reproducibility, the environment is fixed as follows: Python 3.12.11, PyTorch 2.8.0+cu126, torchvision 0.23.0+cu126, Albumentations 2.0.8, OpenCV 4.12.0, NumPy 2.0.2, and pandas 2.2.2. The classification preprocessing produces 107,004 paired crops (214,008 PNG files) occupying 12.69 GiB on disk, with manifest CSVs recording absolute paths, labels, and scene membership. Together, these choices (foreground-aware cropping for segmentation, strict paired transforms for classification, and geographically separated splits) are designed to reduce spatial autocorrelation, preserve change geometry, and mitigate domain shift between natural-disaster pretraining and conflict imagery.

3.7 Experimental Design

3.7.1 The segmentation step: Pretraining on xBD

For the building-footprint segmentation task, a ResUNet model (Figure 1) with a ResNet-34 encoder is employed. Encoder-decoder U-Net derivatives remain standard for footprint extraction and post-disaster mapping since skip connections preserve fine spatial detail while deep encoders capture broader context (*Ronneberger, Fischer, & Brox, 2015; Jha, Smedsrud, Riegler, & Johansen, 2019*). Multiple studies report strong performance of U-Net-ResUNet variants for damage-mapping precursors and related segmentation tasks in remote sensing (*e.g., xBD/xView2 and follow-ups; Gupta et al., 2019; Ji et al., 2024; Wu et al., 2021*). A residual backbone offers a stable optimisation and a favourable accuracy-compute trade-off on high resolution imagery. ResNet-34, particularly, provides adequate capacity without the memory and latency cost of deeper encoders, which is important for tiled training and fast iterations (*He et al., 2016*). Although recent change-detection work explores Siamese and transformer designs (*Fang, Li, Shao, & Li, 2021; Wang, Chen, Guo, Yi, & Li, 2023*), the segmentation stage targets single-temporal footprint delineation. ResUNet aligns with established practice and with the xBD ecosystem commonly used to pretrain building extractors. The footprint network is a ResUNet with a ResNet-34 encoder initialized from ImageNet.

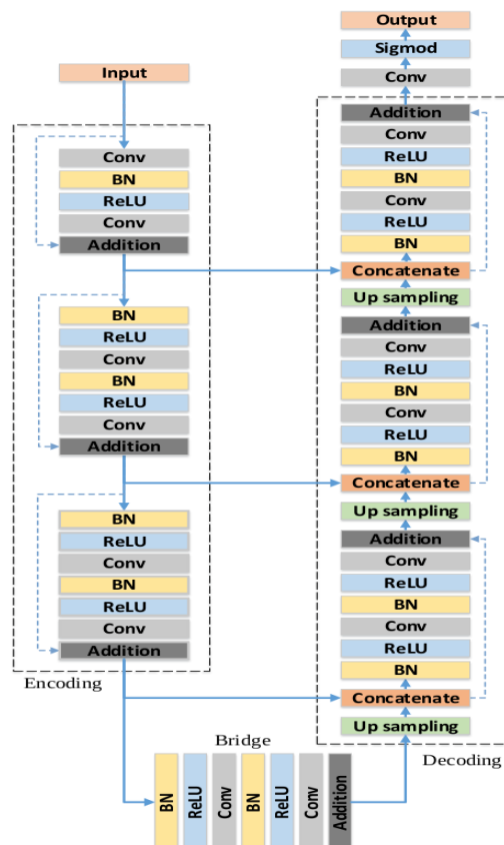


Figure 1 - ResUNet general architecture (Source: N.Tomar, 2021)

Encoder feature maps are taken after the stem (stride 2) and after residual blocks at strides 4, 8, 16, and 32, with channel sizes $\mathbf{c}_0 = 64$, $\mathbf{c}_2 = 64$, $\mathbf{c}_3 = 128$, $\mathbf{c}_4 = 256$, $\mathbf{c}_5 = 512$.

The decoder upsamples by a factor of two at each stage using transposed convolutions and concatenates the corresponding encoder feature map before two 3×3 Conv-BN-ReLU blocks.

Decoder widths are $d_4 = 128$, $d_3 = 64$, $d_2 = 32$, $d_1 = 64$. Bilinear interpolation is applied where needed to align spatial sizes before concatenation and again to project logits back to the input resolution, which helps avoid checkerboard artifacts. A 1×1 projection produces a single logit channel, and a sigmoid is applied at evaluation time. ResNet-34 is preferred over deeper variants to keep memory use and latency compatible with tiled training and inference on high-resolution scenes while retaining sufficient receptive field for urban structures. Batch normalization and ReLU activation functions stabilize the optimisation. The inputs are normalized with ImageNet channel statistics to remain compatible with common encoder initialization; when initialized from ImageNet, this expedites convergence, whereas at inference time the learned checkpoint dominates. The class imbalance issue between building and background is addressed at two levels.

Training optimizes a composite loss that balances region overlap with probabilistic calibration. Initially, the main loss function combines weighted binary cross-entropy with a soft Dice loss term, a combination shown to improve minority-class overlap in imbalanced segmentation. Training minimises a composite objective that balances probabilistic calibration with region overlap (Yeung, Sala, Schönlieb, & Rundo, 2022; Abraham & Khan, 2018). With $\mathbf{p} = \sigma(\mathbf{z})$ denoting the sigmoid of per-pixel logits $\mathbf{z}$, and $\mathbf{y} \in \{0, 1\}$ the ground-truth mask. The weighted Binary Cross-Entropy term is

$$\mathcal{L}_{BCE} = -[\mathbf{w} \mathbf{y} \log \mathbf{p} + (1 - \mathbf{y}) \log (1 - \mathbf{p})]$$

with a global positive weight $\mathbf{w} = \min \left\{ \frac{N_{neg}}{N_{pos}}, 15 \right\}$ estimated once from the training set to avoid unstable gradients under extreme sparsity. The Dice term emphasizes region overlap,

$$\mathcal{L}_{Dice} = 1 - \frac{2\sum \mathbf{p}\mathbf{y} + \epsilon}{\sum \mathbf{p} + \sum \mathbf{y} + \epsilon}$$

and the composite objective is $\mathcal{L} = (1 - \lambda)\mathcal{L}_{BCE} + \lambda\mathcal{L}_{Dice}$ with $\lambda = 0.5$.

Second, at data loading time, the proportion of background-only tiles is controlled through a background sampling probability, ensuring sufficient exposure to empty scenes without collapsing the foreground signal. When a mask-guided crop is unavailable, a deterministic fallback crop maintains sampling robustness. These loader policies complement the loss while leaving labels unchanged and are kept fixed across runs for better compatibility.

To prevent spatial leakage from overlapping tiles, a held-out validation split formed at the scene level is implemented. During validation, a scalar decision threshold $\mathbf{t} \in [0.1, 0.9]$ on the sigmoid output is swept to maximize mean IoU. The resulting $\mathbf{t}$ is then fixed for the final test evaluation to avoid optimistic bias. Reported metrics include IoU, Dice, precision, recall, and accuracy, aggregated over pixels. IoU and Dice are emphasized because they correlate with footprint usability and are a standard for segmentation benchmarks in remote sensing (Gupta et al., 2019; Ji et al., 2024). Qualitative checks de-normalize batches and overlay predictions to confirm that outputs align with roof extents and edges, rather than shadows or repetitive textures. Overall, this configuration (ResUNet with a residual encoder, a BCE + Dice composite loss with dataset-level positive weighting, loader-level balance controls, and validation-driven threshold selection) follows the prevailing template in the literature for reliable and computationally efficient footprint segmentation in building damage-mapping

pipelines (Gupta et al., 2019; Fang et al., 2021; Wang et al., 2023; Ji et al., 2024; Wu et al., 2021).

3.7.2 The Segmentation Step: fine-tuning on damage_assessment_ukraine

Fine-tuning adapts the footprint segmentation network to the “damage_assessment_ukraine” corpus. Scenes are partitioned by original folds to prevent leakage, with training folds ranging from fold_2 to fold_9, validation on fold_1 and fold_4, and held-out test on fold_0. Tiles are processed at 1024x1024 pixels to provide a better feature context, since the corpus is much smaller than the xBD one.

The model is architecturally the same ResUNet with a ResNet-34 encoder initialized from an xBD checkpoint trained for generic building extraction, so the strides, channel sizes and batch-normalization ReLU block widths are identical to the xBD-pretrained version (He et al., 2016; Ronneberger et al., 2015).

The optimisation follows a two-stage schedule designed for stable transfer learning. When initializing from the xBD checkpoint, the encoder is frozen for the first two epochs to allow the decoder and output layer to adapt. The full network is then unfrozen for joint optimisation. AdamW optimizer is used with learning rate 1×10^{-4} and weight decay 5×10^{-5} (Loshchilov & Hutter, 2019). To address the instability that could be generated from large tiles on small batches (batch_n = 4), mixed-precision training with gradient scaling is enabled, and gradient norms are clipped at 1.0. Early stopping monitors the best validation IoU obtained via a threshold sweep, and halts training after eight epochs without improvement. The checkpoint with the highest swept-IoU is kept (Micikevicius et al., 2018; Pascanu et al., 2013). The loss function deployed is the same as the xBD-pretrained model, a combination of binary-cross-entropy with logits and a soft Dice term.

Validation reports IoU, Dice, precision, recall and accuracy at a fixed threshold of 0.5 for continuity with prior logs, and performs a narrowed threshold sweep over [0.75,0.95] with 41 steps to select the operating point that maximizes mean IoU on the validation split. For robustness to small false positives, an optional post-processing pass is applied during validation at the swept threshold: connected components smaller than 64 pixels are removed and a single morphological closing with a 3×3 elliptical kernel smooths building edges (Gonzalez & Woods, 2018; Soille, 2003). The same swept threshold is then used for final testing, and metrics are accumulated over the entire split to provide stable estimates.

3.7.3 The Classification Step: pretraining on xBD

This stage addresses the assignment of damage severity to individual buildings by comparing their pre- and post-event appearance. The framework follows the four-level taxonomy common in humanitarian assessment: undamaged, minor damage, major damage, destroyed. These categories are ordinal and deceptively subtle: boundaries between adjacent classes are often subjective, and satellite viewpoints compress three-dimensional cues into textures, shadows and roof patterns that can be difficult to separate. The modelling challenge, therefore, is not only multi-class prediction but the reliable discrimination of fine-grained changes under varying illumination, viewing geometry and clutter. Two change-aware CNN families are considered: a single-stream early-fusion model that processes concatenated pre/post inputs, and a two-stream Siamese model with shared encoders and a late-fusion head. Early fusion provides a compact baseline but tends to blur temporal roles before any explicit comparison, whereas Siamese encoders preserve temporal structure and generally improve sensitivity to subtle or

localized changes in remote-sensing change detection (Fang et al., 2021; Wang et al., 2023). Accordingly, the expectation is that the Siamese model attains higher macro-F1, with early fusion serving as a strong, efficient reference. Training uses xBD, currently the largest publicly available benchmark for this task, with >850,000 annotated buildings across multiple disaster types and world regions. Its diversity is an asset for pretraining but introduces domain shift when models are later applied to conflict-driven damage, where debris morphology, fire patterns and access damage can differ from natural disasters. Each instance is a paired 256×256 crop centred on the target building. This resolution is a deliberate compromise: smaller crops risk losing contextual cues (e.g., debris fields, neighbouring roof collapses, cast shadows); larger crops reduce batch sizes and destabilise optimisation under the fixed VRAM budgets. To ensure rigorous evaluation, splits are formed at the scene level with approximately 70%/15%/15% allocation to training/validation/test. Scene-level splitting avoids the leakage that arises when nearby buildings from the same neighbourhood appear across splits; such leakage can inflate scores by encouraging memorisation of local textures rather than learning generalisable indicators of damage. Fixed seeds are used for all random operations (splits, initialisation, sampling), enabling like-for-like comparison between architectural variants.

The experimental design contrasts two “philosophies” of temporal processing:

- **Early Fusion** (Figure 2) treats change detection as a single perceptual task from the first layer by concatenating pre- and post-event images into a six-channel input. The hypothesis is that optimal filters for change emerge naturally when both temporal states are processed jointly from the outset.

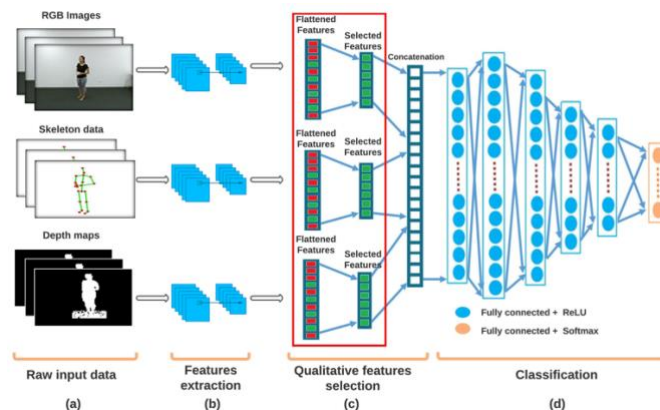


Figure 2 - Early Fusion Net architecture (Source: Boulahia et. al, 2021)

- **Siamese U-Net** (Figure 3) processing keeps the temporal streams separate, allowing each state to develop its own representation with shared weights before an explicit comparison at the fusion head. The premise is that preserving temporal structure improves sensitivity to subtle, localised change and offers robustness under clutter.

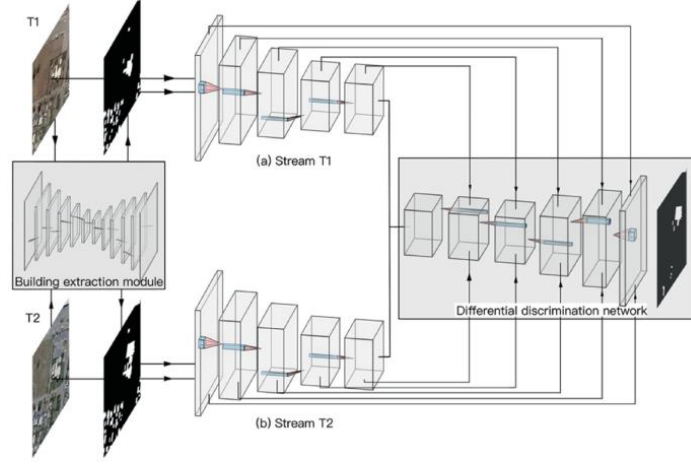


Figure 3 - Siamese U-Net architecture (Source: Song et. al, 2023)

Early fusion is computationally efficient (one encoder pass) but may dilute temporal cues, whereas a Siamese encoder with shared weights tends to yield stronger change sensitivity. Both variants use a ResNet-50 backbone to balance capacity and throughput (He et al., 2016). Accordingly, EarlyFusionNet is retained as a baseline for comparison, and the Siamese U-Net is treated as the main candidate in this research (Table 1 shows the architectural comparison of the two change detection models).

Component	Early Fusion	Siamese
Backbone	ResNet-50	ResNet-50 (shared weights)
Input channels	6 (RGB_pre $\oplus$ RGB_post optionally 9 with post-pre)	3 $\times$ 2 (dual stream)
First conv	3 $\rightarrow$ 6 channel inflation: replicate pretrained 3-ch kernels across channel groups and scale by 1/chunks (2 for 6-ch; 3 for 9-ch) to preserve fan-in	Standard 3-channel
Fusion	Input-level (concatenate at channels)	Late fusion: concat(f_{pre} , f_{post} , $ f_{post}-f_{pre} $, $f_{pre}\odot f_{post}$)
Pooling + head	GAP $\rightarrow$ MLP (Linear $\rightarrow$ ReLU $\rightarrow$ Dropout 0.40 $\rightarrow$ Linear)	GAP $\times$ 2 $\rightarrow$ MLP (Linear $\rightarrow$ ReLU $\rightarrow$ Dropout 0.40 $\rightarrow$ Linear)
Params ($\approx$)	$\sim$ 25.6M	$\sim$ 25.6M (weights shared across streams)
FLOPs ($\approx$, 224 ²)	$\sim$ 4.1 G	$\sim$ 8.2 G (two passes)

Table 1 - Architectural choices for the classification stage in the two-stage pipeline

The Siamese classifier processes pre- and post-event images through weight-sharing encoders to produce descriptors f_{pre} and f_{post} . After global average pooling, the fusion vector $\mathbf{z}$ encodes complementary signals such as the state (raw descriptors), the change magnitude (absolute difference) and co-activation (elementwise product), which is later fed to a compact MLP head.

$$\mathbf{z} = [f_{pre}; f_{post}; |f_{post} - f_{pre}|; f_{post} \odot f_{pre}]$$

The training framework addresses severe class imbalance through three complementary mechanisms operating at different stages of the optimisation pipeline. The objective employs

Focal Loss (Lin et al., 2017) to dynamically modulate gradient contributions based on prediction confidence:

$$\mathcal{L}_{focal} = -\alpha_y (1 - p_t)^\gamma \log p_t$$

where p_t denotes the predicted probability for true class y , and the focusing parameter γ controls the degree of down-weighting for well-classified examples. During pretraining, $\gamma=1.5$ aggressively suppresses easy examples to accelerate minority class learning. Fine-tuning reduces this to $\gamma=1.3$ to maintain gradient stability once reasonable features emerge. This adaptive weighting ensures computational resources concentrate on informative examples near decision boundaries.

Long-tailed class frequencies are a defining feature of damage assessment: undamaged buildings are abundant, while destroyed examples are scarce. Naïvely weighting by the inverse of class frequency can over-correct, tiny classes receive extremely large weights, leading to unstable gradients. The effective-number approach (Cui et al., 2019) addresses this by modelling diminishing returns: the information gain from a class does not grow linearly with the raw count n_k , because many samples within a class are partially redundant.

For this reason, the second component of the mechanism involves handling class weights via effective numbering. Formally, the effective number of a class k with n_k samples is:

$$E_k = \frac{1 - \beta^{n_k}}{1 - \beta}$$

where $0 < \beta < 1$ is a smoothing parameter. β controls how quickly few samples from the same class become redundant.

If β is closer to 1, strong smoothing is applied. Each additional sample adds only a small amount of “new” information; classes with large n_k are down-weighted more aggressively.

If β is closer to 0, the behaviour approaches inverse frequency.

Class weights are then taken as the inverse effective number:

$$w_k = \frac{1}{E_k} = \frac{1 - \beta}{1 - \beta^{n_k}}, \quad \beta = 0.9999,$$

and typically normalized to mean 1 (this way the overall learning rate is not implicitly changed).

In the limit, for a very small n_k and $\beta \approx 1$, $w_k \approx \frac{1}{n_k}$ leading to a likely inverse frequency. For a very large n_k , $w_k \rightarrow (1 - \beta)$, preventing the majority class’s weights to collapse to zero. This formulation naturally tempers the correction for majority classes and stabilises the emphasis on minority classes. In practice it reduces oscillations and overfitting compared with hard class-uniform sampling or raw inverse-frequency weights, especially when paired with Focal Loss, which already re-weights examples by difficulty.

The third component picked for the task is tempered resampling. It is a function that gently flattens the class distribution without going all the way to uniform.

Pure class-uniform resampling can overfit the tail by repeating the same few rare samples; doing nothing leaves the model starved of minority examples. For this reason, this “middle path function” is included in the research.

Let's consider $p(y = k)$ being the empirical class prior in the training split, and $\tau \in [0, 1]$ the tempered exponent. The sampling weights are defined by:

$$s_k \propto p(y = k)^{-\tau}$$

Where if $\tau = 0$ then no resampling is applied.

If $\tau = 1$ then class-uniform sampling is applied.

If $0 < \tau < 1$ then tempered sampling is applied (minority is upweighted, but not to full uniformity).

In mini-batch SGD (stochastic gradient descend), this changes the expected class mix per batch from $p(\cdot)$ towards a tempered distribution:

$$q(k) = \frac{p(k)^{-\tau}}{\sum_j p(j)^{-\tau}}$$

Where k is the specific class for which the new probability is computed.

j is an index that runs through all classes (including k itself).

The denominator $\sum_j p(j)^{-\tau}$ sums the tempered probabilities across all classes.

In this research, the tempering exponent is set to $\tau = 0.5$, a common, pragmatic compromise.

Tempered resampling raises the exposure of minority classes at the data level, so the model sees more hard, rare cases each epoch. Unlike uniform resampling, it preserves some natural frequency, which reduces over-reliance on a handful of rare samples, lowers gradient variance versus hard class balancing, and plays nicely with the proposed loss-level balancing (Focal + effective-number weights). Together with Focal Loss (difficulty-aware) and effective-number weights (class-aware), it forms a three-point control system: data-level exposure (tempered resampling), class-level emphasis (weights), and example-level emphasis (Focal).

Moving to the optimisation and transfer protocol steps, the optimisation framework chosen is AdamW (Loshchilov & Hutter, 2019), a variant of the Adam optimizer that decouples weight decay from gradient-based updates, using a weight decay coefficient of 2×10^{-4} . This decoupling provides more consistent regularisation across different learning rates and prevents the unintended scaling effects that occur with traditional L2 regularisation. The learning rate follows a cosine annealing schedule that smoothly decreases from the initial value to near-zero across the training epochs, avoiding the sudden drops of step-based schedules that can destabilize training in the final phases. Additionally, several engineering optimisations that accelerate training without compromising stability have been implemented. Mixed-precision training leverages modern GPU tensor cores by computing forward passes in FP16 while maintaining FP32 master weights for accurate gradient accumulation, typically yielding 1.5-2 $\times$ speedup with negligible accuracy impact (Micikevicius et al., 2018). The channels-last memory layout aligns tensor storage with hardware memory access patterns, improving cache utilization particularly for convolutional operations. To prevent gradient explosion during early training or when encountering difficult batches, gradient clipping constrains the L2 norm to $\|g\|_2 \leq 1.0$, truncating excessive gradients rather than discarding entire batches (Pascanu et al., 2013). When GPU memory limits prevent an optimal batch size, the gradient accumulation simulates larger batches by aggregating gradients across multiple forward passes before updating weights, maintaining stable optimisation dynamics despite hardware constraints.

3.7.4 The classification step: Fine-tuning on damage_assessment_ukraine

Fine-tuning begins from the best checkpoint obtained during xBD pretraining and adapts the model to the damage_assessment_ukraine dataset. To preserve geographic separation and avoid scene leakage, the original fold structure is retained: fold_1 is used for validation, fold_0 is held out for final testing, and the remaining folds form the training set. All stochastic elements (model initialisation, data loaders, and the sampler) are seeded for reproducibility. In addition, efficiency considerations follow the pretraining stage: automatic mixed precision and a channels-last memory format are enabled throughout, and gradients are clipped at $\|g\|_2 \leq 1.0$ to stabilise updates (Micikevicius et al., 2018; Pascanu et al., 2013). Rather than updating the entire backbone immediately, the optimisation follows a short, two-stage unfreezing schedule designed to mitigate catastrophic forgetting when transferring from natural-disaster imagery (xBD) to conflict damage. The warm-up stage trains only the classification head for two epochs while keeping the encoder frozen; the head learning rate is set to $0.75\times$ the pretraining head rate to respect the strong prior learned on xBD. Once the head has adapted, the encoder is unfrozen for the remaining epochs with a pronounced learning-rate differential: the encoder uses $0.1\times$ the head learning rate, which permits meaningful feature adaptation without excessive representation drift. A cosine-annealing schedule is also applied across both stages, with the horizon reset at unfreezing so that the decay spans the true remainder of training (Loshchilov & Hutter, 2017). In total, training runs for up to fifteen epochs with early stopping (patience five) on validation macro-F1. Batches of 64 samples are used with gradient accumulation to achieve an effective batch size of 128, which improves gradient estimates under the available memory budget.

Addressing label imbalance relies on rebalancing exposure at the data level rather than the loss level. The damage_assessment_ukraine folds exhibit moderate, not extreme, skew: *destroyed* (551) and *minor* (529) are the most common classes, *no-damage* (438) is mid-frequency, and *major* (255) is the clear minority. The batches are drawn using a tempered weighted sampler in which the per-class sampling weight is proportional to $p(class)^{-\tau}$ with $\tau = 0.5$. This choice moves the batch composition towards the tail more gently than pure inverse frequency, increasing minority exposure while retaining some of the natural prevalence of each class and thereby reducing variance. Because the sampler already performs this rebalancing, the loss is intentionally left unweighted; stacking a sampler and class-weighted loss tends to double-count imbalance, which can degrade calibration and over-emphasise rare classes. Within this sampler-only regime, the objective remains focal loss to emphasise examples near decision boundaries (Lin et al., 2017). Given the strong initialisation from xBD, training begins with a moderate focusing parameter $\gamma=1.5$ and may, once validation has stabilised, be swept to $\gamma=2.0$. Class weights inside focal are disabled (as noted above) to avoid piling data- and loss-level corrections on top of one another. In practice this configuration improves minority recall, particularly for the difficult “major” versus “destroyed” boundary, while maintaining stable optimisation with relatively small batch sizes.

Model selection is performed on the validation split without test-time augmentation to provide an unbiased early-stopping signal and checkpoint criterion. The primary selection metric is macro-F1, which treats classes equally and is therefore appropriate under imbalance. The final model (the checkpoint with the best validation macro-F1 on fold_1) is then evaluated on fold_0 with a lightweight flip ensemble (identity, horizontal, vertical, and combined) and logit averaging. Results are reported as overall accuracy and macro-F1 together with per-class precision, recall, and F1, per-scene macro-F1 to characterise geographic robustness, and confusion matrices that make systematic confusions (notably “minor” versus “major”) explicit.

Implementation follows standard, conservative practice. Optimisation uses AdamW with weight decay 2×10^{-4} ; the head learning rate is $0.75 \times$ its pretraining value, while the encoder uses $0.1 \times$ the current head rate during the unfrozen stage. Cosine annealing is parameterised so that T_{max} spans the current stage (head-only or full-unfreeze), the tempered sampler uses $\tau = 0.5$ with “drop_last=True” for stable batch sizes, and stability aids include also AMP, channels-last tensors, gradient clipping, and seeded random number generators. As a small, optional convenience, the final-layer bias may be initialised to the log-priors computed on the training fold to accelerate convergence; this adjustment is benign and does not interact with the sampler-only balancing. These design choices form a conservative but effective protocol suited to domain shift and class imbalance. The two-stage unfreezing preserves useful xBD features while allowing controlled adaptation; extremely small encoder learning rates can stall learning if applied from the outset, and overly large rates erode the pretrained structure, so the 5–10% encoder-to-head ratio is a practical middle ground. Relying on the sampler alone for rebalancing avoids the over-correction that can arise when combining samplers with class-weighted losses. Finally, focal loss with a moderate γ directs learning pressure to the uncertain cases that dominate operational errors in conflict imagery, improving macro-F1 without sacrificing calibration or out-of-region generalisation.

4 Experimental Results and Discussion

4.1 Performance of the proposed Two-Stage Pipeline

The experimental results from a two-stage pipeline designed for assessing building damage are presented in this chapter. The approach combines footprint segmentation with damage classification, and it is assessed using a standardized methodology that prioritizes reproducibility and unbiased comparability across architectures. With a focus on the trade-offs between classification performance and computational efficiency in the damage assessment task, the following results show how effective each strategy is across several evaluation criteria.

4.1.1 Segmentation Stage: ResUNet Performance in Pre-training and Fine-Tuning

The ResUNet model chosen for the segmentation task is trained on 5598 GeoJSON files (2799 pairs) split as 2240 for training, 559 for validation, and 279 for testing. Training proceeded for twenty epochs with AdamW optimizer, cosine annealing, mixed precision and gradient clipping. The composite objective (weighted BCE + soft Dice, $\lambda=0.5$) converged steadily: the training loss decreased steadily and flattened after epoch ~ 15 (Figure 4).

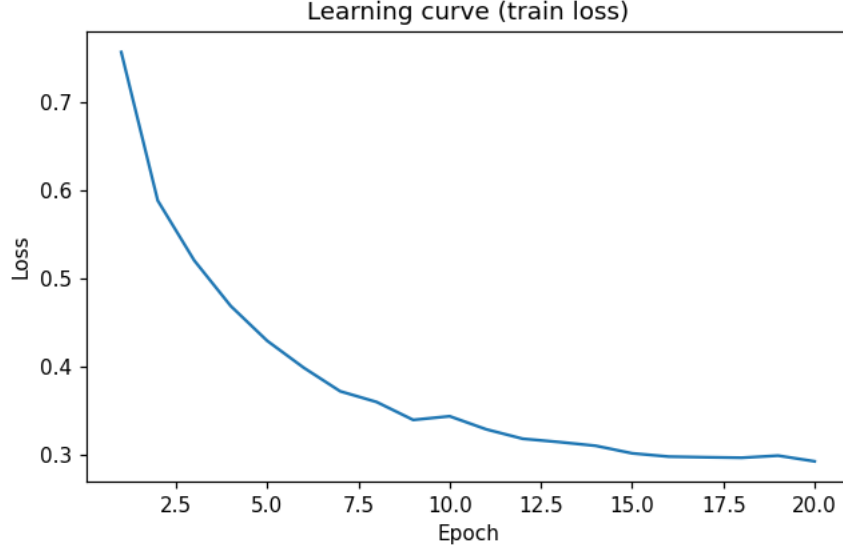


Figure 4 - Loss trend during epochs, Siamese U-Net training phase

Validation metrics surfaced in tandem, with IoU@0.5 (the footprint IoU with its matched ground-truth polygon is ≥ 0.50 .) improving from ~ 0.24 at epoch 1, stabilising around 0.47 at epoch 20; Dice@0.5 (closely related to IoU, slightly rewards overlap on small objects) followed the same trend, reaching ~ 0.64 (Figure 5). The absence of late-epoch divergence between training loss and validation scores suggests no obvious overfitting at this resolution and augmentation strength.

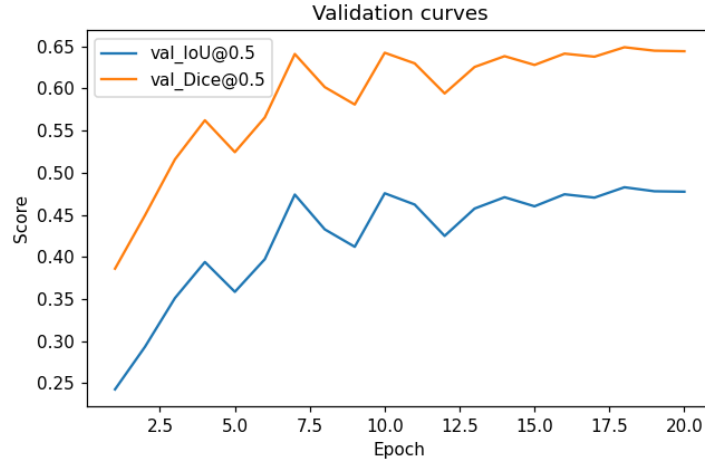


Figure 5 - Validation IoU and Dice @ 0.5 during epochs

The operational point was chosen by sweeping a scalar decision threshold on the sigmoid output due to the threshold-sensitivity of footprint extraction. Since the model's outputs are probabilities, the validation set is swept with a threshold $t \in [0,1]$ in order to maximize mean IoU. To prevent optimistic bias, the selected t is then fixed for testing. Throughout the investigated range, the IoU-threshold curve rises nearly linearly, peaking at $t \approx 0.90$, where the mean IoU on the validation loader reaches around 0.57 (Figure 6).

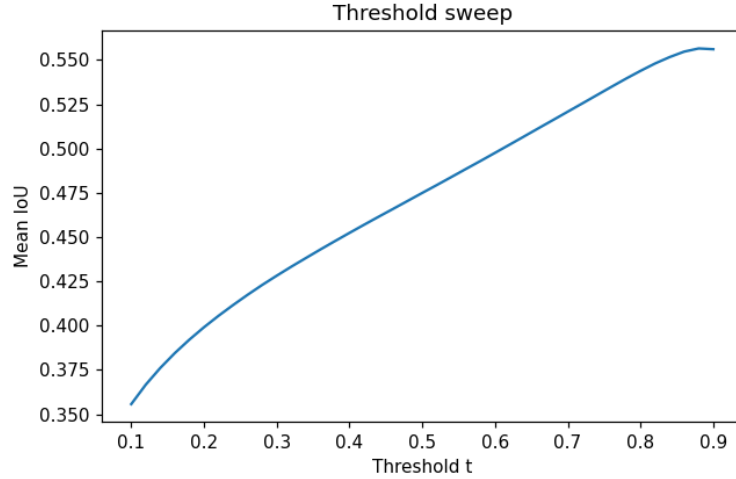


Figure 6 - Threshold sweep (pretraining) for mask binarization: mean IoU as a function of probability threshold t

This preference for a relatively conservative threshold is consistent with the strong background present in overhead imagery, and with the training objective that balances calibration and overlap; higher thresholds reduce small false positives while preserving building cores. Using this swept operating point, the validation yields the metrics report a moderate IoU overlap, where $\sim 57\%$ of the footprint pixels; Dice is consistent with the IoU reported, and precision/recall is near-balanced, with slight recall. This indicates a reliable core detection and damage assessment, with issues on small roofs and boundary edges (Table 2).

Metric 	Validation Set 
IoU	0.568
Dice	0.724
Precision	0.71
Recall	0.739
Accuracy	0.972

Table 2 - Best validation metrics during pretraining (segmentation on xBD)

The quantitative profile aligns with qualitative review. Most buildings, particularly medium and large roofs in residential and commercial areas, are accurately defined at the verified operating point ($t \approx 0.90$), consistent with high recall and Dice scores. Common errors include spurious fragments around bright clutter, parking markers, or vegetation boundaries; partial footprints with truncated edges on small or low-contrast roofs under haze, shadows, or off-nadir conditions. These behaviors explain the precision-recall trade-off and preference for higher thresholds: increasing t suppresses small false positives while preserving building cores but accentuates under-segmentation on marginal structures. Overall, the model provides reliable core footprint recovery with occasional perimeter erosion and removable speckles

(Figure 7, Figure 8) ,an expected U-Net pattern with BCE+Dice training on high-resolution imagery, amendable to light post-processing if needed.

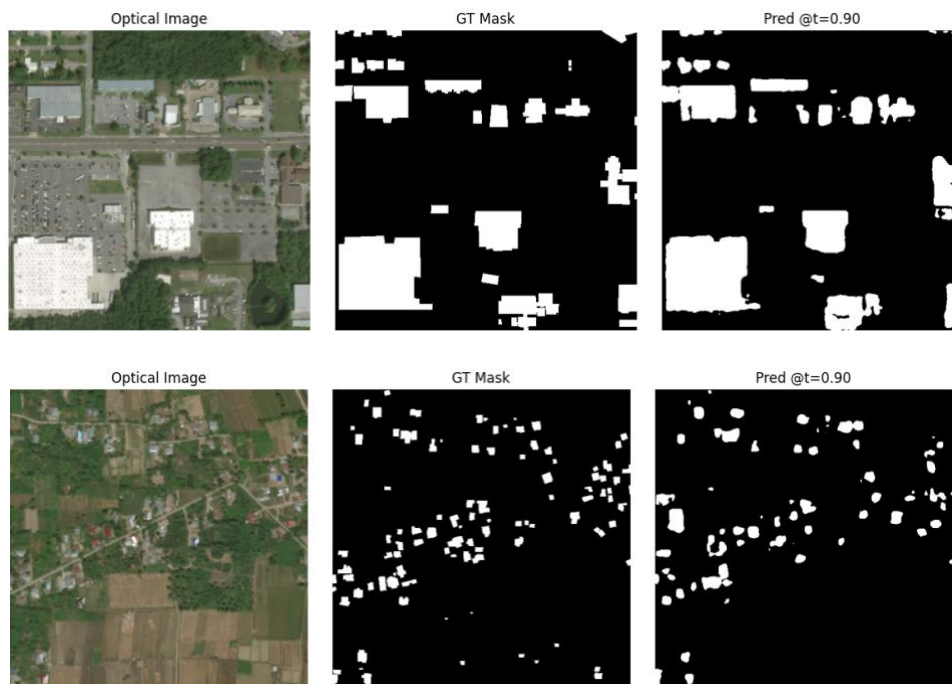


Figure 7 - Optical Images, Respective Ground Truth Masks, and Predicted Masks @ $t=0.90$



Figure 8 - Overlays of Ground Truth borders (Red) and Predicted borders (Green)

Starting with the xBD-pretrained ResUNet-34, the next step involves fine-tuning on "damage_assessment_ukraine" using the same optimisation scheme (AdamW, cosine decay, AMP, gradient clipping). The encoder is frozen for the first two epochs to stabilise the decoder and head before joint optimisation. Given the smaller size of damage_assessment_ukraine and the larger input resolution (1024^2), the longer horizon gave the network more context and opportunities to adapt features without showing signs of overfitting. Training is conducted for up to forty epochs as validation metrics improved until about epoch thirty. Hyperparameters are deliberately a little less conservative than in pretraining: weight decay is increased to 5×10^{-5} (vs. 1×10^{-5}), encouraging crisper solutions under stronger regularisation while retaining the BCE + soft-Dice objective.

The learning dynamics mirror the pretraining step, but plateau later. The training loss decayed smoothly and plateaued after ~ 30 epochs, as shown in Figure 10; validation IoU@0.5 and Dice@0.5 increased steadily before calming. As in the xBD stage, the operating point was chosen by sweeping the sigmoid threshold on the validation split. The sweep again favoured a conservative decision rule, peaking at $t \approx 0.90$, and this threshold was fixed for final testing to avoid optimistic bias (Figure 11), and the training loss decreased continuously and flattened after epoch ~ 30 (Figure 10).

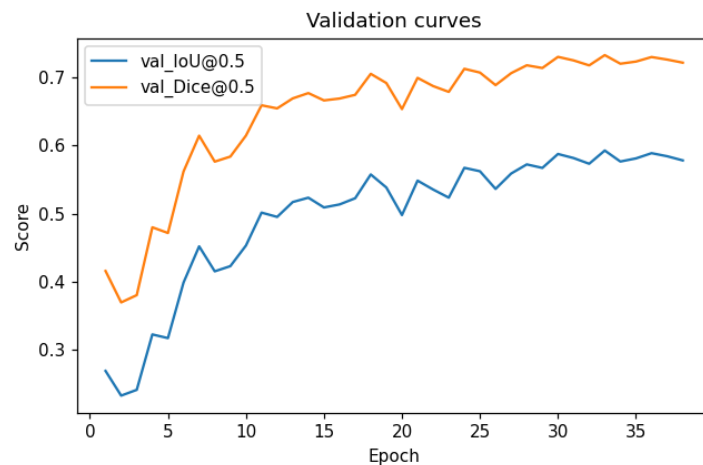


Figure 9 - Validation IoU and Dice, @0.5, Fine-Tuning on damage_assessment_ukraine

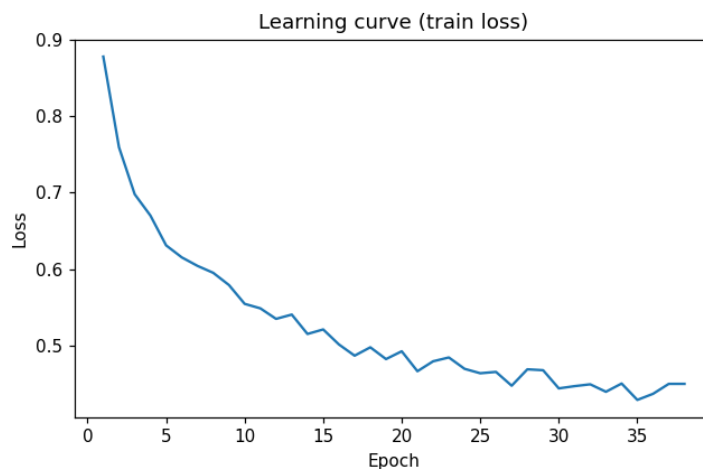


Figure 10 - Loss trend during epochs, Siamese U-Net Fine-Tuning phase

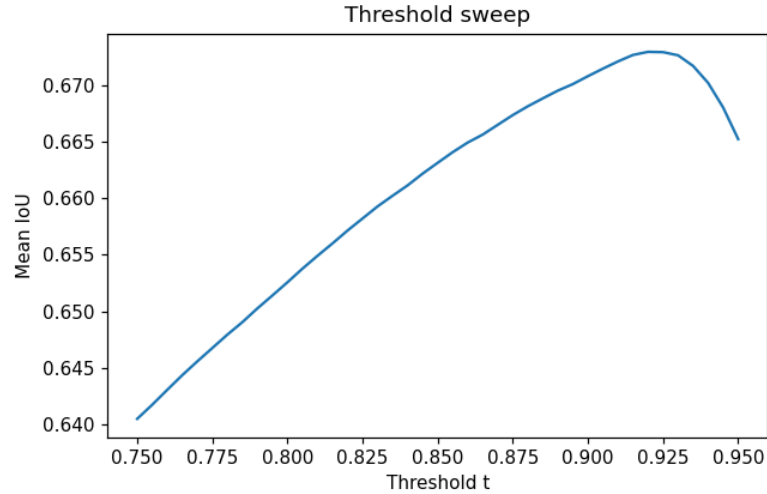


Figure 11 - Threshold sweep (Fine-Tuning) for mask binarization: mean IoU as a function of probability threshold t

The validated operating point demonstrates solid, production-ready performance on held-out data. The model achieves balanced precision-recall with excellent overall accuracy, indicating reliable core detection without heavy bias toward over- or under-detection. The predictable validation behaviour confirms effective suppression of spurious responses while preserving building cores. These results (Table 3) demonstrate a well-calibrated model that prioritizes confident detection over aggressive coverage, making it suitable for damage assessment applications where precision is critical.

Metric ☐	KOlega Test Set ☐
IoU	0.593
Dice	0.744
Precision	0.745
Recall	0.744
Accuracy	0.978

Table 3 - Evaluation metrics on the held-out test set, *damage_assessment_ukraine* Fine-Tuning

The established metrics align with qualitative evaluation. Most buildings in residential and industrial areas are accurately distinguished, with typical errors including rounded corners, edge erosion on small/low-contrast roofs, and occasional bright fragments in congested regions, behaviours explaining the precision-recall balance and conservative threshold choice. Figure 12 presents qualitative examples showing the model tracks building cores well while suppressing background clutter. Success cases include compact residential blocks and larger

industrial roofs with predictions closely matching ground truth. Main error modes surface as expected: small roofs occasionally missed, thin appendages under-segmented, and slight boundary dilation from the conservative threshold. Light post-processing (connected components filtering and morphological closing) smooths edges and minimizes isolated blobs without affecting core footprints.

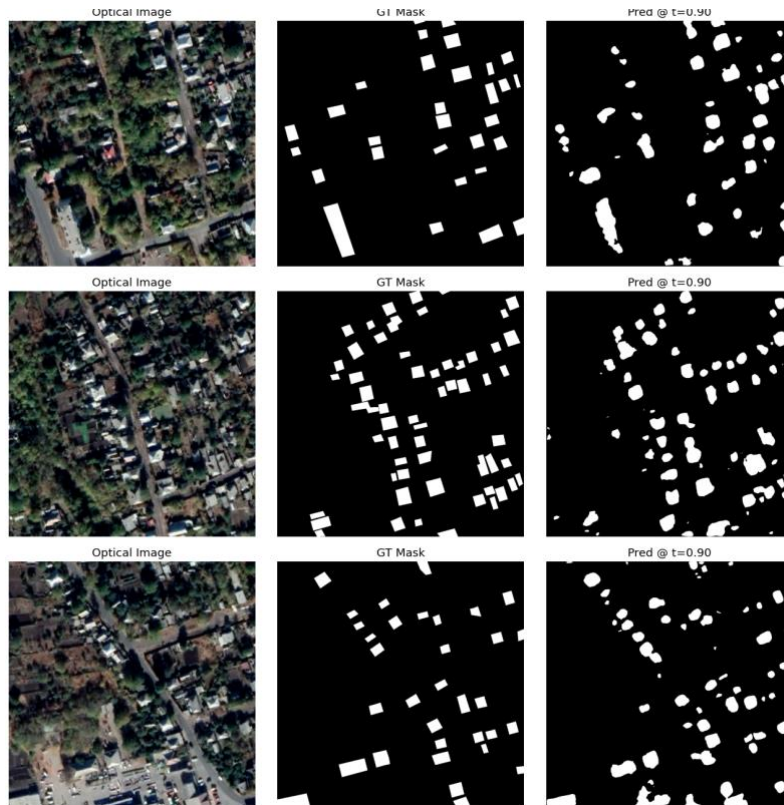


Figure 12 - Qualitative evaluation of Fine-Tuning Segmentation: Optical Image, Ground-Truth mask, and Predicted Mask @ $t=0.90$

To make boundary behaviour explicit, Figure 13 overlays vectorised perimeters on the optical imagery (red = GT, green = prediction). Regions where green extends beyond red indicate false positives; red without green indicates false negatives. The overlays confirm the trade-off seen quantitatively: strong alignment on large, well-defined roofs; irregular misses on small “outbuildings”, and occasional overreach along shadowed edges or vegetation. This visual inspection supports the choice of a precision-first threshold for triage and suggests targeted solutions (e.g., low-threshold ROI sweeps for completeness, light morphological closing, or a minimum-area filter to prune slivers) when higher recall is required.



Figure 13 - Overlays of Ground Truth borders (Red) and Predicted borders (green), Fine-Tuning Stage

Relative to the xBD checkpoint evaluated at its own validated threshold, damage_assessment_ukraine fine-tuning provides a modest gain in overlap (+0.014 IoU, +0.011 Dice) and a clear improvement in precision (+3.4 points) alongside a small decrease in recall (−1.4 points). The longer schedule and slightly stronger regularisation appear well-matched to the smaller, more diverse corpus: learning continues until roughly epoch thirty before plateauing, and there is no evidence of late-epoch overfitting in the curves (Figure 9). In practical application, this adjustment decreases the number of false detections that analysts must manually check, though it slightly raises the chance of overlooking small or faint structures. For the intended purpose of quick assessment and mapping, this approach that prioritizes accuracy is preferable and is validated by the IoU-threshold analysis, which reaches its highest point around $t=0.90$.

4.1.2 The Classification Stage: Pretraining and Fine-Tuning

To compare Early-Fusion and Siamese U-Net change-detection models under identical conditions, paired training crops are derived from full-frame xBD scenes (RGB tiles and JSON labels) (Gupta et al., 2019). Since xBD does not supply building-centred patches, these are constructed as follows. The corpus includes 5,598 PNG images with 5,598 JSON label files, partitioned into 2,799 pre-event and 2,799 post-event images (with 2,799 post-event annotations). All full images are retrieved and indexed in a manifest (manifest.csv), yielding 2,799 scene pairs. To maintain spatial alignment across time, TARGET (post-event) raster masks are used to define instance extents; identically positioned windows are then extracted

from the corresponding pre-event images. JSON-driven crops are excluded because many polygons are missing or too small to define meaningful instances (logged as `json_mask_too_small_or_missing`, 325,574 cases). A small number of TARGET tiles contained no valid components (`target_no_components`, 558 cases). The final dataset contains 107,004 valid building-centred pre/post pairs. The class frequencies (`label_id` $\rightarrow$ class) are:

- **0 $\rightarrow$ no damage:** 73,661
- **1 $\rightarrow$ minor:** 12,244
- **2 $\rightarrow$ major:** 11,784
- **3 $\rightarrow$ destroyed:** 9,315

The distribution exhibits the expected long tail in severe-damage categories, consistent with the xBD dataset's inherent class imbalance where no-damage comprises 49.5% of samples while minor and major damage represent less than 6% each. Accordingly, model comparison uses macro-averaged metrics, macro-F1 as the primary criterion, together with per-class precision, recall, and F1 to capture performance on the rarer "major" and "destroyed" classes. Once the pretraining is complete (capped at 8 epochs, with batches of 577 crops due to computational constraints) the resulted comparative evaluation reveals that the Siamese U-Net architecture demonstrates superior performance relative to the Early-Fusion baseline across all main summary metrics (Table 4). Model performance improved substantially across all metrics. Overall accuracy increased from 0.761 to 0.802, placing our approach competitively against recent CNN ensembles (up to 0.86; *Hernandez et al., 2024*) and matching transformer-based architectures (0.80; *Chen et al., 2022*). The macro-F1 score showed the most significant enhancement, rising from 0.562 to 0.664, matching DeepDamageNet's state-of-the-art 0.66 and far exceeding the xBD baseline of 0.27 (*Gupta et al., 2019*). The weighted-F1 metric increased to 0.809, approaching the xView2 challenge winner's 0.812 (*Weber & Kané, 2020*). These improvements demonstrate robustness even with the dataset's pronounced class imbalance from the dominant no-damage category

Metric	Early-Fusion	Siamese U-Net	Δ
Accuracy	0.761	0.802	0.041
Macro-Precision	0.575	0.639	0.064
Macro-Recall	0.555	0.697	0.142
Macro-F1	0.562	0.664	0.102
Weighted-Precision	0.749	0.82	0.072
Weighted-Recall	0.761	0.802	0.041
Weighted-F1	0.753	0.809	0.056

Table 4 - Comparison of Early-Fusion & Siamese U-Net Evaluation Metrics

As Table 5 shows, the Siamese U-Net architecture improved nearly all per-class metrics. The *no-damage* class F1 score increased from 0.873 to 0.900, matching performance reported by Shen et al. (2020) and confirming the Siamese architecture's effectiveness for majority class classification. This improvement stems primarily from enhanced precision (0.856 to 0.933), despite a slight recall decrease (0.890 to 0.870). This pattern indicates the Siamese architecture adopts a more conservative classification approach for no-damage cases, substantially reducing false positives against damaged classes while maintaining adequate sensitivity. These precision-recall trade-offs align with recent multi-step feature fusion studies (Gašparović et al., 2024) in class-imbalanced scenarios." The *minor damage* category showed considerable improvement, with F1 score rising from 0.420 to 0.524. This was driven primarily by increased recall (0.391 to 0.550) and improved precision (0.453 to 0.500). Confusion matrix analysis reveals that misclassifications of *minor damage* as *no damage* dropped from 706 to 366 instances (Tables 6, 7). The *major damage* category exhibited the most pronounced gains, with F1 score increasing from 0.378 to 0.537. Recall improved sharply from 0.329 to 0.582, indicating the Siamese architecture successfully detected many previously missed *major damage* instances. True positives grew from 511 to 904, while precision improved from 0.444 to 0.498. The *destroyed* category's F1 score climbed from 0.578 to 0.697. Both precision (0.548 to 0.625) and recall (0.610 to 0.787) improved substantially. True positives expanded from 879 to 1,133, while false predictions of *destroyed* for *minor damage* cases declined from 236 to 161 instances.

Class / Metric	Early-Fusion (P/R/F1)	Siamese U-Net (P/R/F1)	Δ (P/R/F1)
no-damage	0.856 / 0.890 / 0.873	0.933 / 0.870 / 0.900	" + 0.077 / -0.021 / +0.027 "
minor	0.453 / 0.391 / 0.420	0.500 / 0.550 / 0.524	" + 0.047 / + 0.159 / + 0.104 "
major	0.444 / 0.329 / 0.378	0.498 / 0.582 / 0.537	" + 0.055 / + 0.253 / + 0.159 "
destroyed	0.548 / 0.610 / 0.578	0.625 / 0.787 / 0.697	" + 0.077 / + 0.176 / + 0.119 "

Table 5 - Comparison of per-class metrics

The confusion matrices examination (Table 6, 7) reveals that the Early-Fusion architecture exhibits regular shortcomings in the detection of damaged classes (*minor*, *major*, and *destroyed*), coupled with frequent misclassification between adjacent severity categories. Particularly prevalent are instances where *major damage* is erroneously classified as either *no-damage* or *minor damage*. In contrast, the Siamese model demonstrates substantial improvements in error reduction, successfully converting a significant proportion of previously misclassified instances (false negatives) into correct classifications (true positives) for both *major* and *destroyed* categories. Furthermore, the architecture demonstrates improved discrimination between *minor* and *destroyed* categories. The residual classification errors occur predominantly between the *minor* and *major* damage categories, which reflects the inherent difficulty in distinguishing between these severity levels when analysing overhead imagery.

Actual \ Pred	No-damage	Minor	Major	Destroyed	Row total	Recall
No-damage	10,636	481	464	369	11,950	0.89
Minor	706	688	129	236	1,759	0.3911
Major	673	250	511	120	1,554	0.3288
Destroyed	413	100	48	879	1,440	0.6104

Table 6 - Early-Fusion confusion matrix

Actual \ Pred	No-damage	Minor	Major	Destroyed	Row total	Recall
No-damage	10,390	679	522	359	11,950	0.8695
Minor	366	967	265	161	1,759	0.5497
Major	247	243	904	160	1,554	0.5817
Destroyed	139	45	123	1,133	1,440	0.7868

Table 7 - Siamese U-Net confusion matrix

The empirical evidence shows that maintaining temporal information in separate pathways with shared encoder architectures (the Siamese approach) substantially improves sensitivity to change detection, particularly for severe damage categories (*major* and *destroyed*), while enhancing precision for *no damage* classification. Given the macro-F1 improvement of 0.10 and superior error profile, the Siamese U-Net represents the more robust candidate for operational deployment. Based on these findings, the study selected the Siamese U-Net as the main classifier. To confirm stable training and absence of overfitting, validation learning curves (loss, accuracy, and macro-F1) were examined, with the best validation checkpoint evaluated on the held-out test split. As Figure 14 shows, training loss declined steadily across eight epochs (0.53 to 0.37), while validation loss followed a parallel trajectory without late-epoch rebound, indicating no overfitting. Validation accuracy consistently exceeded training accuracy throughout (0.705 vs 0.630 at epoch 1, persisting through epoch 8). This pattern aligns with robust regularisation and augmentation strategies that make training more challenging than validation, signalling healthy generalisation rather than memorisation

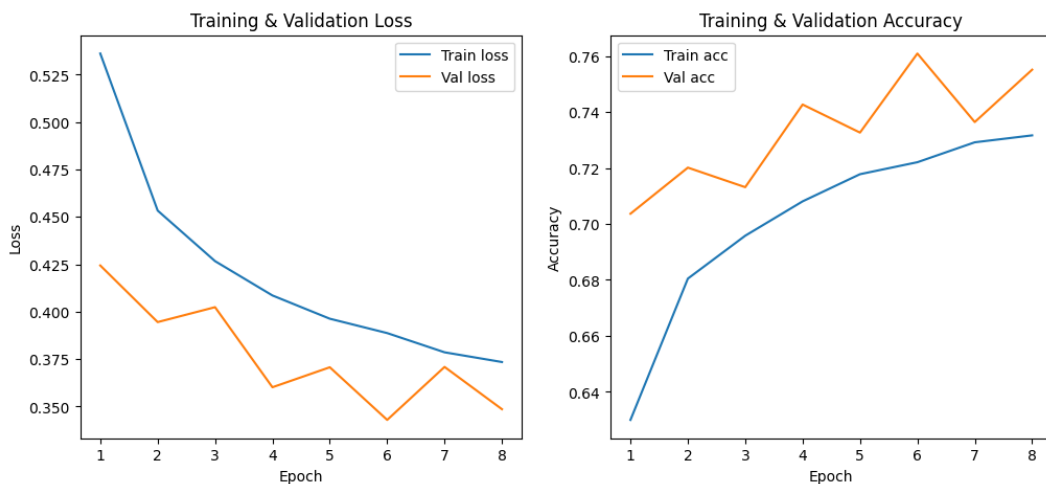


Figure 14 - Training and Validation: Accuracy and Loss during epochs

The primary model-selection metric, macro-F1, increases from approximately 0.575 at epoch 1 to 0.637 at validation epoch 8 (Figure 15). Collectively, the curves (Figure 14, 15) indicate continued improvement with diminishing returns; further training may yield only small incremental gains and increased risk of overfitting.

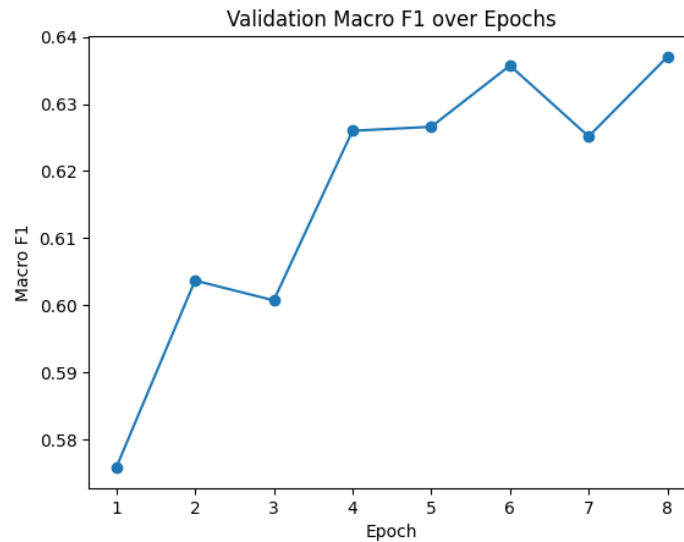


Figure 15 - Pretraining Macro-F1 during Validation

Utilising the checkpoint corresponding to the highest validation macro-F1 score, an evaluation on the unseen test split confirms out-of-sample robustness, achieving an accuracy of 0.802 and a macro-F1 score of 0.664 (Table 8). Class-wise diagnostics, documented through precision, recall, and F1 metrics alongside confusion matrices, reveal the most substantial improvements in the less frequent minor, major, and destroyed categories compared to the early-fusion baseline (Table 9). These findings align with the architectural hypothesis that the Siamese design more effectively preserves temporal information critical for damage assessment (Durnov, 2019). There is to note that the metrics are computed on the unseen test split using the best checkpoint chosen by validation Macro-F1; test-time evaluation uses a light flip-TTA with logit averaging. The same best weights will be reused for the downstream fine-tuning experiments.

Overall Test Set Metrics	Value
Accuracy	0.802
Macro-F1	0.664
Cross-entropy loss	0.306
Best validation Macro-F1 (for reference)	0.637

Table 8 - Evaluation metrics of the held-out test set

Class	Precision	Recall	F1-score	Support
No-damage	0.933	0.87	0.9	11,950
Minor	0.5	0.55	0.524	1,759
Major	0.498	0.582	0.537	1,554
Destroyed	0.625	0.787	0.697	1,440

Table 9 - Per-class evaluation metric for the held-out test set

Fine-tuning adapts the pretrained Siamese U-Net to the `damage_assessment_ukraine` corpus using a strictly geographically disjoint protocol. Fold 1 serves as validation, fold 0 remains sealed for testing, and remaining folds constitute the training set. The four-level taxonomy shows moderate class imbalance: training contains 438, 529, 255, and 551 instances; validation has 50, 66, 27, and 80; test has 49, 76, 30, and 68. Major damage is the scarcest class, while destroyed is well represented. Starting from the best pretraining checkpoint, a short warm-up trains the head before unfreezing the encoder layer-wise with a smaller learning rate to limit drift from xBD features. Automatic mixed precision, channels-last tensors, and gradient clipping stabilise optimisation under small batches. Model selection uses validation macro-F1 without TTA, with best performance at epoch 14 (macro-F1 ≈ 0.571). The learning curves indicate stable transfer. Training and validation loss fall sharply after unfreezing and then stabilise around 0.55 without a late-epoch rebound, which would otherwise signal overfitting (Figure 16). Accuracy for both splits rises steadily, with validation accuracy oscillating around 0.58–0.63 while tracking the training curve (Figure 16). Most importantly, validation macro-F1 recovers from early volatility and trends upward to 0.571 by epoch 14, suggesting ongoing but diminishing returns (Figure 17); additional epochs might produce incremental gains but were not pursued due to computational constraints.

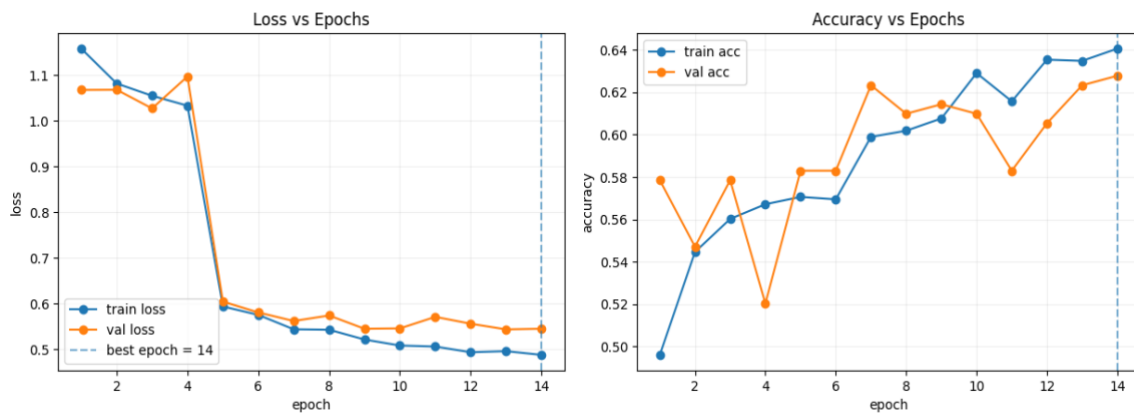


Figure 16 - Training and Validation: Accuracy and Loss during Fine-Tuning

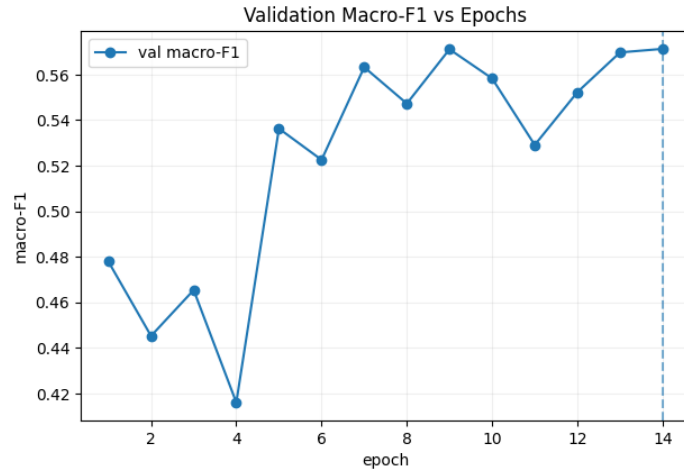


Figure 17 - Validation Macro-F1 during epochs, Fine-Tuning stage

The best validation checkpoint was evaluated on the held-out test fold using light flip-based test-time augmentation (TTA) with logit averaging. Performance results (Table 10) reveal a clear hierarchy among damage classes: the model performs best on the *destroyed* category, showing strong detection capabilities for severe damage. The *no-damage* and *minor* damage classes achieve moderate, comparable performance, while *major* damage proves most challenging to detect, consistent with its scarcity in the training data. This performance pattern aligns with the class distribution challenges noted earlier, where major damage represents the smallest category yet requires nuanced distinction from both minor damage and destroyed states

Class	Precision	Recall	F1	Support
No-damage	0.568	0.51	0.538	49
Minor	0.591	0.513	0.549	76
Major	0.36	0.3	0.327	30
Destroyed	0.648	0.838	0.731	68
Macro / overall	—	—	0.54	223

Table 10 - Per-class evaluation metrics of the test set+TTA

The confusion matrix structure confirms a consistent identification of destroyed buildings; when mistakes occur, they mostly arise from heavily damaged structures being pushed upward into this class. Major damage is the least stable category, leaking both upward to destroyed and downward to minor, functioning as a transitional bucket rather than a crisp class. Minor and no-damage are occasionally swapped, particularly in dense blocks or on small roofs where texture and shadows obscure cues. Mislabels mostly occur between neighbouring categories rather than skipping across the scale, which matches the ordinal label design and the thin pool of authoritative examples at the minor/major boundary (Table 11). Operationally, this profile

supports reliable triage of catastrophic loss, while moderate damage warrants analyst review and benefits from boundary-aware training and tighter calibration

True \ Pred	No-damage	Minor	Major	Destroyed	Row total
No-damage	25	19	1	4	49
Minor	15	39	11	11	76
Major	0	5	9	16	30
Destroyed	4	3	4	57	68
Col total	44	66	25	88	223

Table 11 - Confusion Matrix of the evaluation of test set+TTA

A small post-hoc calibration was applied by searching a constant logit bias for the *major* class, increasing sensitivity to major damage and rebalancing errors without retraining. This adjustment improved held-out test performance to 0.578 accuracy and 0.563 macro-F1. As Table 12 shows, the macro-F1 gain stems primarily from dramatically improved major-class recall (doubling from ~0.30 to ~0.67), while *destroyed* performance remains strong and *no-damage* stays stable. The *minor* class F1 decreases modestly, an expected trade-off when prioritising serious damage detection. In operational triage, this trade-off is acceptable since missing substantial damage carries greater consequences than over-flagging borderline cases. The post-calibration confusion matrix appears in Table 13, while Figure 18 compares per-class F1 before (TTA only) and after calibration (TTA + bias).

TTA + Bias				
Class	Precision	Recall	F1	Support
No-damage	0.581	0.51	0.543	49
Minor	0.604	0.421	0.496	76
Major	0.364	0.667	0.471	30
Destroyed	0.708	0.75	0.729	68
Macro / overall	—	—	0.563	223

Table 12 -Evaluation metrics of the test set+TTA+bias

True \ Pred	No-damage	Minor	Major	Destroyed	Row total
No-damage	25	17	3	4	49
Minor	14	32	22	8	76
Major	0	1	20	9	30
Destroyed	4	3	10	51	68
Col total	43	53	55	72	223

Table 13 - Confusion matrix of the evaluation of test set+TTA+bias

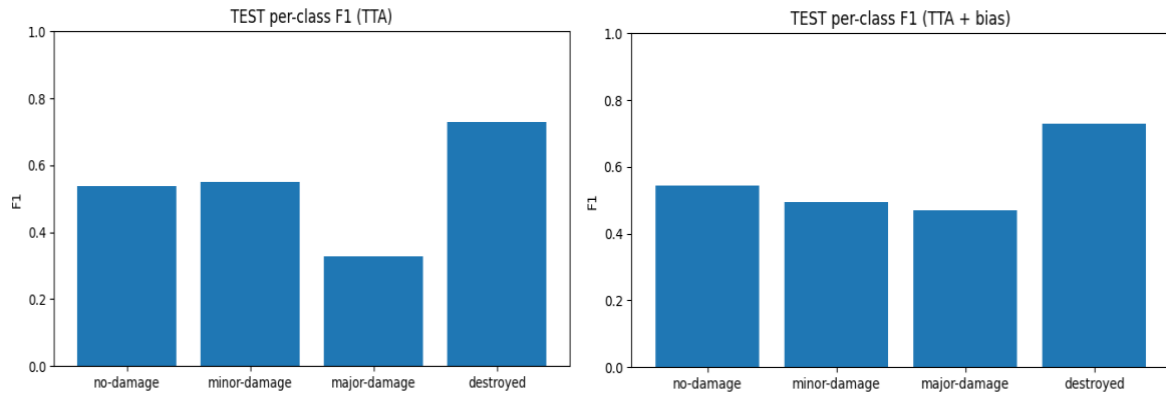


Figure 18 - Visualization of the influence of bias

In summary, fine-tuning on `damage_assessment_ukraine` yields validation macro-F1 of ~ 0.571 and held-out test macro-F1 of 0.540 with TTA, improving to 0.563 with lightweight bias adjustment targeting major damage. The principal error mode remains minor-major misclassification, unsurprising given the subtle ordinal boundary and overhead imaging geometry. Destroyed buildings are robustly detected, while no-damage precision and recall remain balanced after transfer.

To examine whether these quantitative gains manifest in realistic imagery, Figure 19 presents qualitative evaluation of both footprint segmentation and damage classification using the best experimental checkpoints (segmentation: swept-threshold model; classification: fine-tuned Siamese model). This dual assessment verifies error patterns implied by the metrics and surfaces operational failure modes that aggregate scores may mask. Pre/post image pairs covering identical footprints were randomly selected across diverse eastern Ukraine locations, testing operational robustness on non-processed imagery. Scene-level and location-stratified selection avoided local bias; all panels are cropped and anonymised per disclosure policy. Semi-transparent overlays show predicted building outlines (thin borders) and damage levels using an intuitive colour gradient: green (no-damage), yellow (minor), orange (major), and red (destroyed). While results are consistent on the curated `damage_assessment_ukraine` data, performance degrades on raw, previously unseen pre/post imagery from platforms such as Google Earth. The segmenter tends to overlook large, damaged complexes and to thin out dense residential blocks, absorbing them into background. The classifier also becomes unstable,

sometimes elevating intact areas to *destroyed* and, conversely, softening obvious collapses to *minor* or, even worse, *no-damage*. Considering these shortcomings, the pipeline is dependable on well-matched imagery but requires explicit adaptation, normalisation, domain alignment, stronger multi-scale features, and tighter co-registration, before use on raw vendor mosaics.



Figure 19 - Qualitative examples of footprint segmentation and building-wise damage classification on unprocessed pre/post satellite imagery.

4.2 Analysis and Discussion

4.2.1 Addressing the Research Questions

The first research question states whether buildings can be reliably found in unseen satellite imagery. The xBD-pretrained ResUNet-34 achieves $\text{IoU} = 0.568$ and $\text{Dice} = 0.724$ on the best validation checkpoint. Following on fine-tuning, performance on the held-out `damage_assessment_ukraine` test set improves to $\text{IoU} = 0.593$ and $\text{Dice} = 0.744$, with pixel accuracy rising from 0.972 to 0.978 and precision/recall shifting from 0.71/0.739 (pretraining) to 0.745/0.744 (fine-tuning). The improvements show a more balanced error profile and improved foreground delineation in the conflict domain. In line with previous studies (e.g. *Yuan et al., 2021*), the footprint extractor is precise enough to support downstream instance extraction and per-building classification, especially for solitary or medium-sized structures, even though the study’s aspirational aim (≥ 0.65 mean IoU) has not yet been reached. As discussed previously, residual errors concentrate in dense urban fabric with many small roofs and in large multi-block complexes; consequently, IoU/Dice remain the primary, honest indicators of footprint quality (pixel accuracy is inflated by background dominance) (*Muller et al., 2022*). Overall, RQ1 is partially satisfied. The model generalises to unseen scenes and benefits from `damage_assessment_ukraine` adaptation, but further improvement (e.g., higher-resolution tiling, boundary-aware losses, or domain-specific pretraining) is required to reach ≥ 0.65 IoU and to reduce misses in dense urban areas, as demonstrated in recent studies (*Quian et.al 2024; Hu et. al 2024; Appati et. al 2024*)

The second research question concerns whether damage can be graded usefully at the building level. On xBD pretraining, the Siamese classifier substantially outperforms the early-fusion baseline: accuracy rises from 0.761 to 0.802, macro-F1 from 0.562 to 0.664, and weighted-F1 from 0.753 to 0.809. Gains concentrate in the rarer “major” and “destroyed” classes, which are operationally critical. This pattern matches change-detection studies in which weight-sharing Siamese encoders preserve temporal structure and improve sensitivity to subtle, localised differences compared with single-stream concatenation (*Daudt et al., 2018; Fang et al., 2021; Wang et al., 2023*). After fine-tuning on `damage_assessment_ukraine`, validation macro-F1 reaches ≈ 0.571 and held-out test macro-F1 is 0.540 with flip-TTA, increasing to 0.563 after a light prior-aligned bias on the “major” logit. These levels are consistent with recent conflict-specific work that also reports moderate macro-F1 under strong class imbalance and label subjectivity, especially at the “minor-major” boundary (*Risso et al., 2024; Zheng et al., 2024*).

The third research question concerns which design choices most improve reliability. Mechanistically, the results support the hypothesis that temporal separation with shared encoders increases sensitivity to subtle, localised change. The Siamese pathway reduces leakage between pre/post content that can blur cues in early fusion, which is reflected in higher recall for “major” and “destroyed” during pretraining and preserved advantages after fine-tuning. Within training, the results support three mechanism-level takeaways. First, temporal separation with shared encoders is decisive: Siamese processing enables the fusion head to combine state features with absolute differences and elementwise products, which sharpens change cues; Early-Fusion, by contrast, forces the first layers to disentangle state and change simultaneously, and tends to blur temporal roles (*Daudt et al., 2018; Fang et al., 2021*). Second, difficulty-aware optimisation matters: focal loss concentrates learning on ambiguous examples, while effective-number weighting and tempered sampling ($\tau = 0.5$) avoid over-correcting majority classes under long-tailed, ordinal taxonomies (*Cui et al., 2019; Yeung et al., 2022*). Third, the transfer protocol is consequential: a short head-only warm-up followed by

unfreezing with a small encoder-to-head learning-rate ratio preserves useful xBD features while permitting controlled adaptation to `damage_assessment_ukraine`, consistent with domain-adaptation guidance (Long *et al.*, 2015; Sun & Saenko, 2016; Ganin *et al.*, 2016) and conflict fine-tuning reports (Risso *et al.*, 2024; Zhao *et al.*, 2024).

The fourth research question asks whether the system is robust enough for operational use. Two lines of evidence are positive: the footprint extractor generalises across `damage_assessment_ukraine` scenes and supports object-level aggregation, and the classifier’s held-out performance, although below xBD levels, remains informative for damage assessment, especially for “destroyed,” which achieves the highest test F1. These benefits increase when outputs are accompanied by uncertainty indicators and are reviewed by an analyst for borderline “minor vs. major” calls. However, inference on previously unseen, unseeded pre/post Google Earth imagery reveals robustness gaps. The segmenter often misses large, damaged complexes and suppresses dense residential fabric, treating damaged areas as background; the classifier exhibits widespread misassignments, including predicting “minor” for clearly destroyed buildings and “destroyed” for intact structures. From the literature (Mueller *et al.*, 2021; Zheng *et al.*, 2024; Risso *et al.*, 2024), likely drivers include domain shift in radiometry, resampling, and colour/sharpening along the imagery pipeline (off-manifold textures and edges relative to xBD/`damage_assessment_ukraine`), scale and composition effects when moving from foreground-biased crops to heterogeneous full frames without strong multi-scale features, taxonomy/appearance drift between disaster-dominated xBD and conflict-specific damage morphology), and pre/post misregistration and illumination differences, which degrade local change cues (Sundaresan *et al.*, 2007). These failures qualify the pipeline’s current operational readiness: it is reliable within the curated `damage_assessment_ukraine` setting but insufficiently robust in the wild without explicit adaptation.

4.2.2 Theoretical and Practical Implications

The performance improvements demonstrated by the proposed approach carry significant implications for both disaster response theory and operational practice. Beyond the numerical gains in macro-F1 scores, these enhancements directly address critical challenges in rapid damage assessment: reducing both the human cost of missed severe damage and the resource waste from false alarms. Macro-F1 improvements translate into fewer missed severe cases and fewer false alarms in undamaged areas. In pretraining, true positives for “destroyed” increase from 879 to 1,133 while false “destroyed” assignments on “minor” drop notably; “major” recall improves from 0.329 to 0.582. Under fine-tuning, “destroyed” remains the most reliable class (test F1 ≈ 0.73 with TTA), which is consistent with recent conflict studies prioritizing catastrophic damage for rapid response (Mueller *et al.*, 2021; Dietrich *et al.*, 2025). The residual error mode, the confusions between “minor” and “major”, is typical in overhead imagery and is repeatedly documented in inter-annotator studies and xView2 error analyses (Manzini *et al.*, 2025; Wu *et al.*, 2021). The two-stage design, footprint segmentation followed by per-building classification, is now standard, and the choices made here broadly align with the prevailing template. ResUNet-like encoders remain the standard choice for footprint extraction (Ronneberger *et al.*, 2015; Wu *et al.*, 2021), often initialised on ImageNet to stabilise optimisation and speed convergence (He *et al.*, 2016). BCE + Dice and related hybrid objectives are widely reported to improve minority-class overlap in imbalanced segmentation (Abraham & Khan, 2018; Yeung *et al.*, 2022). Post-processing with small morphological operators or CRFs to smooth jagged edges is common and improves topological realism (Krähenbühl & Koltun, 2011; Wu *et al.*, 2021). This approach yields high recall for the most

critical class (“destroyed”), which enables rapid triage maps and PDNA-style roll-ups for emergency coordination (UNOSAT/JRC PDNA, 2010; Mueller et al., 2021; Dietrich et al., 2025). Outputs can be integrated into standard GIS workflows (e.g., QGIS dashboards) to prioritise tasking, cue manual quality-assurance, and focus crowd mapping on hotspots rather than uniform coverage (QGIS Development Team, 2025; QGIS.org, 2025; HOT, 2025; Knoth et al., 2018).

The research makes three principal contributions to the field. Firstly, it is demonstrated that transfer learning from natural disaster imagery (xBD) to conflict damage provides performance gains, with the Siamese architecture's temporal separation proving decisive for change detection sensitivity. Secondly, it is demonstrated that complementary optimisation mechanisms, namely focal loss, effective-number weighting, and tempered sampling, effectively address the severe class imbalance inherent to damage assessment. Thirdly, it has been demonstrated that critical robustness gaps are exposed when models are deployed beyond curated datasets. Performance is shown to degrade substantially on unprocessed imagery due to more prominent domain shift, scale mismatches and conflict-specific damage morphology absent from the training and fine-tuning data. These findings carry important operational implications. The system has been demonstrated to achieve sufficient accuracy for preliminary damage assessment within controlled conditions, particularly for catastrophic damage requiring immediate response. However, persistent confusion between minor and major damage categories, combined with failures on out-of-distribution imagery, necessitates human-in-the-loop validation for operational deployment. The discrepancy between experimental performance and real-world robustness emphasises that current models, while methodologically sound, necessitate substantial adaptation prior to autonomous deployment in active conflict zones.

4.2.3 Limitations of the Research

The credibility of the results rests on sensible experimental controls, but several features limit how far the findings can be taken. Seeds are fixed and splits are made at scene level to reduce leakage, yet outcomes come from a small number of runs and lack confidence intervals across random initialisations. Hyper-parameter search is deliberately narrow to respect the compute budget; only a small set of values for the focusing parameter, learning-rate ratios, and unfreezing strategies is explored. Operating points use a validation-threshold sweep for segmentation and macro-F1-based early stopping for classification. These choices are reasonable, but alternative settings could shift the reported figures slightly. The study also adopts a conservative posture: geographic identifiers are withheld, and outputs are framed as decision support rather than autonomous assessment, establishing appropriate boundaries for responsible development. Generalisation beyond the curated setting is the main challenge. The pipeline transfers knowledge from xBD, dominated by natural disasters across many countries, to `damage_assessment_ukraine`, which contains conflict damage from three Ukrainian locations. This introduces changes in debris patterns, season, illumination and sensor processing. The study partly addresses this by spot-checking random pre/post pairs from several areas along the eastern front, but wider geographic and seasonal coverage would be needed before claiming theatre-wide reliability. Label design adds further uncertainty: the four damage grades are ordinal, and the boundary between minor and major is inherently subjective. `Damage_assessment_ukraine` was annotated by a small team without expert adjudication, so some apparent errors may reflect taxonomy ambiguity rather than model failure. Resource limits shape the design. GPU memory constraints require 256×256 crops, moderate batch sizes and short training schedules. Pretraining is capped at eight epochs (about 3.5 hours end-to-end),

and fine-tuning runs up to fourteen epochs with early stopping. Broader sweeps over architectures, input resolutions and stronger augmentation are therefore out of scope. Within these bounds, the model performs solidly on the internal benchmark: the footprint segmenter reaches IoU 0.593 (Dice 0.744) on the `damage_assessment_ukraine` test split, while the building-wise classifier achieves macro-F1 0.540 with test-time flips, rising to 0.563 after a light prior-aligned bias. These numbers, together with steady learning curves and no late-epoch overfitting, suggest the reported performance is a conservative lower bound. The gap to deployment, however, is clear. Applied to new, unprocessed Google Earth imagery, performance drops sharply. The segmenter sometimes delineates single structures well but often misses large, damaged complexes and dense urban blocks, treating them as background. The classifier shows similar weakness, mixing evident classes, for example, labelling some destroyed buildings as minor and, conversely, some intact buildings as destroyed. Likely causes include vendor-specific image pipelines (colour management, sharpening, compression and orthorectification), scale/context mismatch from crop-based training, and conflict-specific signatures under-represented in xBD. In short, the model travels well within the dataset family on which it is trained and tuned but degrades on out-of-distribution imagery. To reduce this gap, the next steps are clear and tractable. Domain adaptation beyond simple fine-tuning should be added to align feature distributions between sources, including colour and tone normalisation across vendors. Multi-scale encoders or pyramid pooling would allow the classifier to use wider context and handle small buildings in dense neighbourhoods. Incorporating SAR-based change cues would improve resilience under cloud, smoke and variable illumination, and help disambiguate subtle roof damage. Another natural extension can be 3D change detection with LiDAR/terrestrial point clouds (*Xiao et al., 2023*). Finally, the work should continue to follow a harm-minimising posture: figures remain cropped and anonymised; coordinates and city identifiers are withheld; and outputs are presented as decision support with uncertainty, keeping analysts in the loop for high-stakes sites such as hospitals and schools.

5 Conclusion

The present study investigated the development of an automated two-stage deep learning pipeline for building damage assessment in conflict zones, with a particular focus on addressing the critical intelligence gap in the Russia-Ukraine war. The objective of the study was to identify whether AI systems could reliably detect buildings and grade damage levels to support operational decision-making in active conflict environments.

The experimental results demonstrate a degree of success regarding the stated objectives. The segmentation stage achieved IoU of 0.593 and a Dice coefficient of 0.744 on the held-out Ukrainian test set, falling short of the target $\text{IoU} \geq 0.65$ but proving sufficient for the following classification tasks. The Siamese U-Net classifier demonstrated a substantial improvement in performance when compared with the early-fusion baseline, achieving a macro-F1 score of 0.664 during xBD pretraining and 0.563 on Ukrainian conflict imagery following fine-tuning and calibration. While falling short of the aspirational $\text{F1} \geq 0.70$ target, the system exhibited proficiency in detecting destroyed buildings ($\text{F1} \approx 0.73$), the most operationally critical category for humanitarian response. This study makes three contributions. First, transferring knowledge from natural-disaster imagery (xBD) to conflict damage, using a Siamese network that keeps pre- and post-disaster views separate, boosts change-detection sensitivity, especially for major

and destroyed buildings. Second, focal loss, effective-number weighting and tempered sampling together counter severe class imbalance. Third, performance drops on unprocessed images or unseen regions because of domain shifts, scale mismatches and conflict-specific damage patterns absent from the training set. Collectively, these results add to the growing evidence that deep learning can meaningfully support conflict-damage assessment, while still requiring human oversight.

Future research should prioritise three technical interventions. Domain adaptation techniques that extend beyond simple fine-tuning, such as CORAL alignment and adversarial training, have the potential to mitigate performance disparities across imaging vendors and acquisition conditions, and should include colour and tone normalisation across vendors to align feature distributions between sources. Multi-scale architectures with expanded receptive fields, through multi-scale encoders or pyramid pooling, would better accommodate the varied urban morphologies encountered in operational settings and help with small buildings in dense neighbourhoods. The integration of Synthetic Aperture Radar (SAR) imagery has the potential to enhance all-weather capability, improve resilience under cloud, smoke and variable illumination, and help disambiguate subtle roof damage; another improvement could be 3D change detection with LiDAR/terrestrial point clouds. The ethical dimensions of this work demand continued attention. While the technology demonstrates potential for accelerating damage assessment and prioritising humanitarian response, its limitations, particularly systematic misclassifications and vendor-specific fragility, could misdirect resources if deployed prematurely. The work maintains a harm-minimising direction: figures are cropped and anonymised, coordinates and city identifiers are withheld, and outputs are presented with uncertainty as decision support, keeping analysts in the loop for high-stakes sites such as hospitals and schools. These advances mitigate current deployment difficulties but also sharpen the ethical imperative: without them, premature use could cause a misallocation of scarce resources. The path from validation to operations remains steep; the two-stage pipeline is a solid base, yet theatre-wide robustness will require sustained work on domain adaptation, multimodal fusion, and calibration to conflict-specific signatures. Until these advances are addressed, the technology's role must remain supplementary, augmenting rather than replacing human expertise in these high-stakes humanitarian applications.

6 References

- Abraham, N., & Khan, N. M. (2018). *A novel focal Tversky loss function with improved Attention U-Net for lesion segmentation*. arXiv. <https://doi.org/10.48550/arXiv.1810.07842>
- Aimaiti, Y., Sanon, C., Koch, M., Baise, L. G., & Moaveni, B. (2022). War-related building damage assessment in Kyiv, Ukraine, using Sentinel-1 radar and Sentinel-2 optical images. *Remote Sensing*, 14(24), 6239. <https://doi.org/10.3390/rs14246239>
- Al Rahrooh, A., Garlid, A. O., Bartlett, K., Coons, W., Petousis, P., Hsu, W., & Bui, A. A. T. (2024). Towards a framework for interoperability and reproducibility of predictive models. *Journal of Biomedical Informatics*, 149, 104551. <https://doi.org/10.1016/j.jbi.2023.104551>
- Alisjahbana, I., Li, J., Strong, B., & Zhang, Y. (2024, May 8). DeepDamageNet: A two-step deep-learning model for multi-disaster building damage segmentation and classification using satellite imagery [Preprint]. arXiv. <https://doi.org/10.48550/arXiv.2405.04800>
- Alparone, L., Arienzo, A., & Lombardini, F. (2024). Improved coherent processing of synthetic aperture radar data through speckle whitening of single-look complex images. *Remote Sensing*, 16(16), 2955. <https://doi.org/10.3390/rs16162955>
- Amnesty International. (2016, October 13). Digital volunteers to expose Darfur human rights violations. <https://www.amnesty.org/en/latest/press-release/2016/10/digital-volunteers-to-expose-darfur-human-rights-violations/>
- APXML. (n.d.). Segmentation evaluation metrics. In *CNNs for computer vision* (Chap. 4). Retrieved September 4, 2025, from <https://apxml.com/courses/cnns-for-computer-vision/chapter-4-image-segmentation-techniques/segmentation-evaluation-metrics>
- Appati, J. K., Azuponga, N. A., Boante, L. M., & Mensah, J. A. (2024). A multiattention ResUNet and modified U-Net architecture for liver tumor segmentation. *Applied Computational Intelligence and Soft Computing*, 2024, 8365349. <https://doi.org/10.1155/2024/8365349>
- Ashraf, S., Brabyn, L., & Hicks, B. J. (2013a). Introducing contrast and luminance normalisation to improve the quality of subtractive resolution merge technique. *International Journal of Image and Data Fusion*, 4(3), 230–251. <https://doi.org/10.1080/19479832.2013.782072>
- Ashraf, S., Brabyn, L., & Hicks, B. J. (2013b). Landsat-based early warning system to detect the destruction of villages in Darfur, Sudan. *Remote Sensing of Environment*, 133, 39–49. <https://doi.org/10.1016/j.rse.2013.01.025>
- Avtar, R., Kouser, A., Kumar, A., Singh, D., Misra, P., Gupta, A., Yunus, A. P., Kumar, P., Johnson, B., Dasgupta, R., Sahu, N., & Rimba, A. B. (2021). Remote sensing for international peace and security: Its role and implications. *Remote Sensing*, 13, 439. <https://doi.org/10.3390/rs13030439>

- Baatz, M., & Schäpe, A. (2000). Multiresolution segmentation—An optimization approach for high quality multi-scale image segmentation. In J. Strobl, T. Blaschke, & G. Griesebner (Eds.), *Angewandte Geographische Informationsverarbeitung XII* (pp. 12–23). Wichmann.
- Ballinger, O. (2024a). Open-access battle damage detection via pixel-wise t-test on Sentinel-1 imagery. *arXiv*. <https://doi.org/10.48550/arXiv.2405.06323>
- Ballinger, O. (2024b). *Pixel-Wise T-Test (PWTT) for battle damage detection* [Source code]. GitHub. <https://github.com/ollieballinger/pwtt>
- Benz, U. C., Hofmann, P., Willhauck, G., Lingenfelder, I., & Heynen, M. (2004). Multi-resolution, object-oriented image analysis—A new paradigm for remote sensing. *ISPRS Journal of Photogrammetry and Remote Sensing*, 58(3–4), 239–258.
- Blaschke, T. (2010). Object based image analysis for remote sensing. *ISPRS Journal of Photogrammetry and Remote Sensing*, 65(1), 2–16.
- Blaschke, T. (2013). Object based image analysis: A new paradigm in remote sensing? In *American Society for Photogrammetry and Remote Sensing Annual Conference (ASPRS 2013)* (pp. 36–43). American Society for Photogrammetry and Remote Sensing. <https://www.researchstudio.at/wp-content/uploads/2020/05/Blaschke-T.-2013-Object-Based-Image-Analysis.pdf>
- Boulahia, S. Y., Amamra, A., Madi, M. R., & Daikh, S. (2021). Early, intermediate and late fusion strategies for robust deep learning-based multimodal action recognition. *Machine Vision and Applications*, 32, 121. <https://doi.org/10.1007/s00138-021-01249-8>
- Braik, A., & Koliou, M. (2024). Automated building damage assessment and large-scale mapping by integrating satellite imagery, GIS, and deep learning. *Computer-Aided Civil and Infrastructure Engineering*, 39, 2389–2404. <https://doi.org/10.1111/mice.13197>
- Braithwaite, A., Gohdes, A., Gurr, T. R., Liedl, P., Muchlinski, D., Rød, E. G., & Weidmann, N. B. (2023). Toward the remote monitoring of armed conflicts. *PNAS Nexus*, 2(6), pgad181. <https://doi.org/10.1093/pnasnexus/pgad181>
- Buslaev, A., Iglovikov, V. I., Khvedchenya, E., Parinov, A., Druzhinin, M., & Kalinin, A. A. (2020). Albumentations: Fast and flexible image augmentations. *Information*, 11(2), 125. <https://doi.org/10.3390/info11020125>
- Bui, T.-A., & Le, T.-T. (n.d.). Cloud-aware SAR fusion for enhanced optical sensing in space missions [Preprint]. Institute of Aerospace and System Engineering, National Taipei University of Technology; Department of Information Technology, Ho Chi Minh City University of Education.
- Center for Strategic and International Studies. (2024, November 12). Understanding the military AI ecosystem of Ukraine. <https://www.csis.org>
- Chan, D., Gambini, J., & Frery, A. C. (2022). Entropy-based non-local means filter for single-look SAR speckle reduction. *Remote Sensing*, 14(3), 509. <https://doi.org/10.3390/rs14030509>

Chen, Z., Yu, B., Yang, C., Zhou, Y., Yao, S., Qian, X., Wang, C., Wu, B., & Wu, J. (2021). An extended time series (2000–2018) of global NPP-VIIRS-like nighttime light data from a cross-sensor calibration. *Earth System Science Data*, 13, 889–906. <https://doi.org/10.5194/essd-13-889-2021>

Chen, J., Zhang, Z., Xie, X., Wu, Y., Xie, T., Bai, Y., & Xie, E. (2022). Building damage assessment based on Siamese hierarchical transformer framework. *Mathematics*, 10(11), 1898. <https://doi.org/10.3390/math10111898>

Cui, Y., Jia, M., Lin, T.-Y., Song, Y., & Belongie, S. (2019). Class-balanced loss based on effective number of samples. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 9268–9277). <https://doi.org/10.1109/CVPR.2019.00949>

Delcourt, C. J. F., Combee, A., Izbicki, B., Mack, M. C., Maximov, T., Petrov, R., Rogers, B. M., Scholten, R. C., Shestakova, T. A., van Wees, D., & Veraverbeke, S. (2021). Evaluating the differenced normalized burn ratio for assessing fire severity using Sentinel-2 imagery in Northeast Siberian larch forests. *Remote Sensing*, 13(12), 2311. <https://doi.org/10.3390/rs13122311>

Deng, L., & Wang, Y. (2022). Post-disaster building damage assessment based on improved U-Net. *Scientific Reports*, 12, 15862. <https://doi.org/10.1038/s41598-022-20114-w>

DeVries, T., & Taylor, G. W. (2018). Improved regularization of convolutional neural networks with Cutout. *arXiv*. <https://doi.org/10.48550/arXiv.1708.04552>

Dietrich, O., Peters, T., Sainte Fare Garnot, V., et al. (2025). An open-source tool for mapping war destruction at scale in Ukraine using Sentinel-1 time series. *Communications Earth & Environment*, 6, 215. <https://doi.org/10.1038/s43247-025-02183-7>

Durnov, V. (2020). *xview2 1st place solution* [Computer software]. GitHub. Retrieved September 15, 2025, from https://github.com/vdurnov/xview2_1st_place_solution.

European Commission Joint Research Centre (JRC), United Nations Institute for Training and Research (UNITAR) Operational Satellite Applications Programme (UNOSAT), Global Facility for Disaster Reduction and Recovery (GFDRR), & Centre National d'Information Géo-Spatial (CNIGS). (2010, March 11). *Building damage assessment report: Haiti earthquake 12 January 2010—Post-Disaster Needs Assessment and Recovery Framework (PDNA)* (Version 3.0). https://lfestorage.s3.us-east-2.amazonaws.com/images/2010_01_12_haiti/pdfs/PDNA_damage_assessment_report_v03-1.pdf

Fang, S., Li, K., Shao, J., & Li, Z. (2021). SNUNet-CD: A densely connected Siamese network for change detection of VHR images. *IEEE Geoscience and Remote Sensing Letters*. Advance online publication. <https://doi.org/10.1109/LGRS.2021.3056416>

Fechner, T., Wilhelm, D., & Kray, C. (2015). Ethermap. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems (CHI '15)* (pp. 3583–3592). <https://doi.org/10.1145/2702123.2702536>

- Fisher, S., Abdi, D. I., Ludin, J., Smith, R., Williams, S., & Williams, S. (2000). *Working with conflict: Skills and strategies for action*. Zed Books.
- Ganin, Y., Ustinova, E., Ajakan, H., Germain, P., Larochelle, H., Laviolette, F., Marchand, M., & Lempitsky, V. (2016). Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 17(59), 1–35.
- Gašparović, M., Dobrinić, D., & Medak, D. (2024). Multi-step feature fusion for natural disaster damage assessment on satellite images. *arXiv preprint arXiv:2410.21901*. <https://doi.org/10.48550/arXiv.2410.21901>
- Ghandour, A. J., & Jezzini, A. A. (2018). Post-war building damage detection. *Proceedings*, 2(7), 359. <https://doi.org/10.3390/ecrs-2-05172>
- Giglio, L., Schroeder, W., & Justice, C. O. (2016). The Collection 6 MODIS active-fire detection algorithm and fire products. *Remote Sensing of Environment*, 178, 31–41. <https://doi.org/10.1016/j.rse.2016.02.054>
- GitHub. (n.d.). *prs-eth/ukraine-damage-mapping-tool*. <https://github.com/prs-eth/ukraine-damage-mapping-tool>
- Google. (2025a, June 4). *Google Earth End User Additional Terms of Service*. https://www.google.com/help/terms_maps-earth/ (Retrieved September 11, 2025)
- Google. (2025b, June 4). *Google Maps End User Additional Terms of Service*. https://www.google.com/help/terms_maps/ (Retrieved September 11, 2025)
- Google. (n.d.). *Google Enterprise Geo Master License: Google Earth Pro Terms of Service Addendum*. <https://earth.google.com/intl/en/licensepro.html> (Retrieved September 11, 2025)
- Gonzalez, R. C., & Woods, R. E. (2018). *Digital image processing* (4th ed.). Pearson.
- Gupta, R., Hosfelt, R., Sajeev, S., Patel, N., Goodman, B., Doshi, J., Heim, E., Choset, H., & Gaston, M. (2019). xBD: A dataset for assessing building damage from satellite imagery. *arXiv*. <https://doi.org/10.48550/arXiv.1911.09296>
- Hay, G. J., & Castilla, G. (2008). Geographic Object-Based Image Analysis (GEOBIA): A new name for a new discipline. In T. Blaschke, S. Lang, & G. J. Hay (Eds.), *Object-Based Image Analysis* (pp. 75–89). Springer.
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 770–778). IEEE. <https://doi.org/10.1109/CVPR.2016.90>
- Hernandez, I., Gao, J., & Hossain, M. N. (2024). DeepDamageNet: A two-step deep-learning model for multi-disaster building damage segmentation and classification using satellite imagery. *arXiv preprint arXiv:2405.04800*. <https://doi.org/10.48550/arXiv.2405.04800>

Hu, D., & Al Shafian, S. (2024, March). Segmentation and tracking of moving objects on dynamic construction sites. In *Construction Research Congress 2024*. American Society of Civil Engineers. <https://doi.org/10.1061/9780784485262.007>

Hu, Y., Janowicz, K., & Couclelis, H. (2017). Prioritizing disaster mapping tasks for online volunteers based on information value theory. *Geographical Analysis*, 49(2), 175–198. <https://doi.org/10.1111/gean.12117>

Hugging Face. (n.d.). *Hugging Face Hub*. <https://huggingface.co>

Humanitarian OpenStreetMap Team. (2025, May 27). OpenStreetMap contributors needed to map Khartoum ahead of new radar satellite imagery. *HOT*. <https://www.hotosm.org/updates/openstreetmap-contributors-needed-to-map-khartoum-ahead-of-new-radar-satellite-imagery-2025/>

Huo, C., Chen, K., Zhang, S., Wang, Z., Yan, H., Shen, J., Hong, Y., Qi, G., Fang, H., & Wang, Z. (2025). When remote sensing meets foundation model: A survey and beyond. *Remote Sensing*, 17(2), 179. <https://doi.org/10.3390/rs17020179>

International Committee of the Red Cross. (2024). International humanitarian law and the challenges of contemporary armed conflicts. *International Review of the Red Cross*, 106(927), 1357–1450. <https://doi.org/10.1017/S181638312400064X>

Ji, Y., Wu, W., Wan, G., Zhao, Y., Wang, W., Yin, H., Tian, Z., & Liu, S. (2024). Segment anything model-based building footprint extraction for residential complex spatial assessment using LiDAR data and very high-resolution imagery. *Remote Sensing*, 16(14), 2661. <https://doi.org/10.3390/rs16142661>

Jha, D., Smedsrud, P. H., Riegler, M. A., & Johansen, D. (2019). ResUNet++: An advanced architecture for medical image segmentation. *arXiv*. <https://doi.org/10.48550/arXiv.1911.07069>

Jiang, W., He, G., Long, T., & Liu, H. (2017). Ongoing conflict makes Yemen dark: From the perspective of nighttime light. *Remote Sensing*, 9(8), 798. <https://doi.org/10.3390/rs9080798>

Kaur, N., Lee, C. C., Mostafavi, A., & Mahdavi-Amiri, A. (2023). Large-scale building damage assessment using a novel hierarchical transformer architecture on satellite images. *Computer-Aided Civil and Infrastructure Engineering*, 38(15), 2072-2091. <https://doi.org/10.1111/mice.12981>

Kerle, N. (2011, March 23). Remote sensing based post-disaster damage mapping – Ready for a collaborative approach? *Earthzine*. <https://earthzine.org/remote-sensing-based-post-disaster-damage-mapping-ready-for-a-collaborative-approach/>

Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., Xiao, T., Whitehead, S., Berg, A. C., Lo, W.-Y., Dollár, P., & Girshick, R. (2023). Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (pp. 4015–4026).

- Knoth, C., Slimani, S., Appel, M., & Pebesma, E. (2018). Combining automatic and manual image analysis in a web-mapping application for collaborative conflict damage assessment. *Applied Geography*, 97, 25–34. <https://doi.org/10.1016/j.apgeog.2018.05.016>
- Krähenbühl, P., & Koltun, V. (2011). Efficient inference in fully connected CRFs with Gaussian edge potentials. In *Advances in Neural Information Processing Systems (NeurIPS 2011)* (pp. 109–117).
- Lin, N. N. N. (2024, March 1). *Deep learning-based documentation of conflict destroyed buildings in Myanmar with VHR satellite images* (Master's thesis, Universidade Nova de Lisboa). NIMS – MSc Dissertations Geospatial Technologies. <http://hdl.handle.net/10362/165511>
- Lin, T.-Y., Goyal, P., Girshick, R., He, K., & Dollár, P. (2017). Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)* (pp. 2980–2988). <https://doi.org/10.1109/ICCV.2017.324>
- Long, M., Cao, Y., Wang, J., & Jordan, M. I. (2015). Learning transferable features with deep adaptation networks. In *Proceedings of the 32nd International Conference on Machine Learning* (pp. 97–105). PMLR.
- Loshchilov, I., & Hutter, F. (2017). SGDR: Stochastic gradient descent with warm restarts. *International Conference on Learning Representations (ICLR)*. <https://doi.org/10.48550/arXiv.1608.03983>
- Loshchilov, I., & Hutter, F. (2019). Decoupled weight decay regularization. *International Conference on Learning Representations (ICLR)*. <https://doi.org/10.48550/arXiv.1711.05101>
- Ma, X., Zhang, X., Pun, M.-O., & Huang, B. (2025). A unified framework with multimodal fine-tuning for remote sensing semantic segmentation. *IEEE Transactions on Geoscience and Remote Sensing*. Advance online publication. <https://doi.org/10.1109/TGRS.2025.3585238>
- Malila, W. A. (1980, June 3–6). Change vector analysis: An approach for detecting forest changes with Landsat. In *Proceedings of the 6th Annual Symposium on Machine Processing of Remotely Sensed Data* (pp. 326–335). Purdue University/LARS. https://docs.lib.purdue.edu/lars_symp/385/
- Manzini, T., Perali, P., Tripathi, J., & Murphy, R. R. (2025). Now you see it, now you don't: Damage label agreement in drone & satellite post-disaster imagery. In *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency (FAccT '25)* (pp. 1998–2008). Association for Computing Machinery. <https://doi.org/10.1145/3715275.3732135>
- Martakis, P., Reuland, Y., Stavridis, A., & Chatzi, E. (2022). Fusing damage-sensitive features and domain adaptation towards robust damage classification in real buildings. *Soil Dynamics and Earthquake Engineering*, 164, 107739. <https://doi.org/10.1016/j.soildyn.2022.107739>

- Marx, A. J., & Loboda, T. V. (2013). Landsat-based early warning system to detect the destruction of villages in Darfur, Sudan. *Remote Sensing of Environment*, 136, 126–134. <https://doi.org/10.1016/j.rse.2013.05.006>
- Marx, A., Windisch, R., Shaver, A., Martínez, L. R., Schemer, L., & Rost, N. (2019). Detecting village burnings with high-cadence smallsats: A case-study in the Rakhine State of Myanmar. *Political Geography*, 69, 32–40. <https://doi.org/10.1016/j.polgeo.2018.12.009>
- Micikevicius, P., Narang, S., Alben, J., et al. (2018). Mixed precision training. *International Conference on Learning Representations (ICLR)*. <https://doi.org/10.48550/arXiv.1710.03740>
- Mueller, H., Groeger, A., Hersh, J., Matraga, A., & Serrat, J. (2021). Monitoring war destruction from space using machine learning. *Proceedings of the National Academy of Sciences of the United States of America*, 118(23), e2025400118. <https://doi.org/10.1073/pnas.2025400118>
- Nielsen, A. A., Conradsen, K., & Simpson, J. J. (1998). Multivariate alteration detection (MAD) and MAF postprocessing in multispectral, bitemporal image data: New approaches to change detection studies. *Remote Sensing of Environment*, 64(1), 1–19. [https://doi.org/10.1016/S0034-4257\(97\)00162-4](https://doi.org/10.1016/S0034-4257(97)00162-4)
- NASA Earthdata. (2024). *NASA and IBM openly release geospatial AI foundation model for NASA Earth observation data*. NASA. <https://www.earthdata.nasa.gov/news/nasa-ibm-openly-release-geospatial-ai-foundation-model-nasa-earth-observation-data>
- O'Donnell, T. M. (2024). *Quantitative validation of NASA ARIA damage proxy maps* [Doctoral dissertation, University of California, Los Angeles].
- Otsu, N. (1979). A threshold selection method from gray-level histograms. *IEEE Transactions on Systems, Man, and Cybernetics*, 9, 62–66.
- Page, E. S. (1954). Continuous inspection schemes. *Biometrika*, 41(1–2), 100–115.
- Pascanu, R., Mikolov, T., & Bengio, Y. (2013). On the difficulty of training recurrent neural networks. In *Proceedings of the 30th International Conference on Machine Learning (ICML)* (pp. 1310–1318).
- QGIS Development Team. (2025). *QGIS Geographic Information System* (Version 3.44) [Computer software]. QGIS Association. <https://www.qgis.org/>
- QGIS.org. (2025). *QGIS User Guide/Manual* (Version 3.40). QGIS Association. https://docs.qgis.org/3.40/en/docs/user_manual/index.html (Retrieved September 11, 2025)
- Qian, L., Wen, C., Li, Y., Hu, Z., Zhou, X., Xia, X., & Kim, S.-H. (2024). Multi-scale context UNet-like network with redesigned skip connections for medical image segmentation. *Computer Methods and Programs in Biomedicine*, 243, 107885. <https://doi.org/10.1016/j.cmpb.2023.107885>
- Risso, M., Goffi, A., Motetti, B. A., Burrello, A., Bove, J. B., Macii, E., Poncino, M., Jahier Pagliari, D., & Maffei, G. (2024, October 7). Building damage assessment in conflict zones:

A deep learning approach using geospatial sub-meter resolution data [Preprint]. *arXiv*. <https://doi.org/10.48550/arXiv.2410.04802>

Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. In N. Navab, J. Hornegger, W. M. Wells, & A. F. Frangi (Eds.), *Medical image computing and computer-assisted intervention – MICCAI 2015* (Lecture Notes in Computer Science, Vol. 9351, pp. 234–241). Springer. https://doi.org/10.1007/978-3-319-24574-4_28

Sabir, A., Khati, U., Lavalley, M., & Srivastava, H. S. (2024). Adapting CuSUM algorithm for site-specific forest conditions to detect tropical deforestation. *Remote Sensing*, 16(20), 3871. <https://doi.org/10.3390/rs16203871>

Scher, C., & Van Den Hoek, J. (2025a). Active InSAR monitoring of building damage in Gaza during the Israel–Hammas War. *arXiv*. <https://doi.org/10.48550/arXiv.2506.14730>

Scher, C., & Van Den Hoek, J. (2025b). Nationwide conflict damage mapping with interferometric synthetic aperture radar: A study of the 2022 Russia–Ukraine conflict. *Science of Remote Sensing*, 11, 100217. <https://doi.org/10.1016/j.srs.2025.100217>

Shen, Y., Wei, Y., Zhang, H., Rui, X., Li, B., & Wang, J. (2024). Unsupervised change detection in HR remote sensing imagery based on local histogram similarity and progressive Otsu. *Remote Sensing*, 16(8), 1357. <https://doi.org/10.3390/rs16081357>

Shen, Y., Zhu, S., Yang, T., Chen, C., Pan, D., Chen, J., Xiao, L., & Du, Q. (2020). Pyramid pooling module-based semi-Siamese network: A benchmark model for assessing building damage from xBD satellite imagery datasets. *Remote Sensing*, 12(24), 4055. <https://doi.org/10.3390/rs12244055>

Soille, P. (2003). *Morphological image analysis: Principles and applications* (2nd ed.). Springer.

Song, Y., Jing, Z., & Li, M. (2023). Siamese U-net with attention mechanism for building change detection in high-resolution remote sensing images. In Z. Jing & D. Strelets (Eds.), *Proceedings of the International Conference on Aerospace System Science and Engineering 2021* (Lecture Notes in Electrical Engineering, Vol. 849, pp. 487–503). Springer. https://doi.org/10.1007/978-981-16-8154-7_37

Stewart, R., & Hinds, G. (2023, October 24). Algorithms of war: The use of artificial intelligence in decision making in armed conflict. *Humanitarian Law & Policy* (ICRC) Blog. <https://blogs.icrc.org/law-and-policy/2023/10/24/algorithms-of-war-use-of-artificial-intelligence-decision-making-armed-conflict/>

Sundaresan, A., Varshney, P. K., & Arora, M. K. (2007). Robustness of change detection algorithms in the presence of registration errors. *Photogrammetric Engineering & Remote Sensing*, 73(4), 375–383. <https://doi.org/10.14358/PERS.73.4.375>

Sun, B., & Saenko, K. (2016). Deep CORAL: Correlation alignment for deep domain adaptation. In *Computer Vision – ECCV 2016 Workshops* (pp. 443–450). Springer.

Tomar, N. (2021, February 6). What is ResUNet. *Idiot Developer*.

<https://idiotdeveloper.com/what-is-resunet/>

United Nations Institute for Training and Research. (2014, March 24). UNOSAT analysis tells story of devastation in South Sudan city. <https://unitar.org/about/news-stories/news/unosat-analysis-tells-story-devastation-south-sudan-city>

United Nations Institute for Training and Research. (n.d.). WHS Aleppo.

<https://unitar.org/sustainable-development-goals/satellite-analysis-and-applied-research/whs-aleppo>

UNOSAT. (2006, September 29). Satellite identification of damage in Beirut, Lebanon.

ReliefWeb. <https://reliefweb.int/map/lebanon/satellite-identification-damage-beirut-lebanon-29-sep-2006>

Wang, D., Chen, X., Guo, N., Yi, H., & Li, Y. (2023). STCD: Efficient Siamese transformers-based change detection method for remote sensing images. *Geo-spatial Information Science*. Advance online publication.

<https://doi.org/10.1080/10095020.2022.2157762>

Wang, G., Li, Z., & Chen, Y. (2025). An overview of industrial image segmentation using deep learning models. *Intelligence & Robotics*, 5(1), 143–180.

<https://doi.org/10.20517/ir.2025.09>

Wang, X., Li, Y., Lin, C., Liu, Y., & Geng, S. (2021). Building damage detection based on multi-source adversarial domain adaptation. *Journal of Applied Remote Sensing*, 15(3), 036503. <https://doi.org/10.1117/1.JRS.15.036503>

Wang, Y., Li, Y., & Zhang, S. (2025). Automatic detection of war-destroyed buildings from high-resolution remote sensing images. *Remote Sensing*, 17(3), 509.

<https://doi.org/10.3390/rs17030509>

Wang, Z., Wu, C., Zhang, F., & Xia, J. (2025). Constructing an extensible building damage dataset via semi-supervised fine-tuning across 12 natural disasters. *Journal of Remote Sensing*, 5, 0733. <https://doi.org/10.34133/remotesensing.0733>

Weber, E., & Kané, H. (2020). Building disaster damage assessment in satellite imagery with multi-temporal fusion. *International Conference on Learning Representations (ICLR) Workshop on AI for Earth Sciences*. <https://doi.org/10.48550/arXiv.2004.05525>

Wu, C., Zhang, F., Xia, J., Xu, Y., Li, G., Xie, J., Du, Z., & Liu, R. (2021). Building damage detection using U-Net with attention mechanism from pre- and post-disaster remote sensing datasets. *Remote Sensing*, 13(5), 905. <https://doi.org/10.3390/rs13050905>

Xiao, W., Cao, H., Tang, M., Zhang, Z., & Chen, N. (2023). 3D urban object change detection from aerial and terrestrial point clouds: A review. *International Journal of Applied Earth Observation and Geoinformation*, 118, 103258.

<https://doi.org/10.1016/j.jag.2023.103258>

Yale Humanitarian Research Lab. (n.d.). Yale Humanitarian Research Lab monitoring conditions in Sudan. *Yale School of Public Health*. <https://ysph.yale.edu/news-article/yale-humanitarian-research-lab-monitoring-conditions-in-sudan/>

Yeung, M., Sala, E., Schönlieb, C.-B., & Rundo, L. (2022). Unified focal loss: Generalising Dice and cross entropy-based losses to handle class imbalanced medical image segmentation. *Computerized Medical Imaging and Graphics*, 95, 102026. <https://doi.org/10.1016/j.compmedimag.2021.102026>

Yin, X.-X., Jian, Y., Shen, J., Wu, J., Zhang, Y., & Wang, W. (2023). Focal boundary Dice: Improved breast tumor segmentation from MRI scan. *Journal of Cancer*, 14(5), 717–736. <https://doi.org/10.7150/jca.82592>

Zhang, W., Zhao, R., Yao, Y., Wan, Y., Wu, P., Li, J., Li, Y., & Zhang, Y. (2025, February 3). Multi-resolution SAR and optical remote sensing image registration methods: A review, datasets, and future perspectives. *arXiv*. <https://doi.org/10.48550/arXiv.2502.01002>

Zhao, C., Cao, X., Chen, X., & Cui, X. (2022). A consistent and corrected nighttime light dataset (CCNL 1992–2013) from DMSP-OLS data. *Scientific Data*, 9, 424.

Zhao, Q., Tan, X., Meng, Q., Gao, L., Zhang, L., Lei, X., ... Zhu, M. (2024). Satellite and AI monitoring humanitarian crises in the Gaza Strip during the early stage of Israeli–Palestinian conflict. *International Journal of Digital Earth*, 17(1). <https://doi.org/10.1080/17538947.2024.2430678>

Zhao, X., & Morikawa, S. (2024). Rapid assessment of large-scale urban destruction in conflict zones using hypergraph-based visual-structural machine learning. *Journal of Engineering Research*. Advance online publication. <https://doi.org/10.1016/j.jer.2024.08.006>

Zheng, Z., Zhong, Y., Zhang, L., Burke, M., Lobell, D. B., & Ermon, S. (2024). Towards transferable building damage assessment via unsupervised single-temporal change adaptation. *Remote Sensing of Environment*, 315, 114416. <https://doi.org/10.1016/j.rse.2024.114416>

Zitzlsberger, G., & Podhorányi, M. (2024). Monitoring of urban changes with multi-modal Sentinel-1 and 2 data in Mariupol, Ukraine, in 2022/23. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 17, 5245–5265. <https://doi.org/10.1109/JSTARS.2024.3362688>

7 Appendix A



Certificate of Ethics Review

Project title: AI-Enhanced Damage Assessment from Satellite Imagery in the Russia-Ukraine Conflict

Name:	Adamo Bovolenta	User ID:	2069775	Application date:	03/06/2025 14:41:41	ER Number:	TETHIC-2025-111203
--------------	-----------------	-----------------	---------	--------------------------	------------------------	-------------------	--------------------

You must download your referral certificate, print a copy and keep it as a record of this review.

The FEC representative(s) for the **School of Computing** is/are [Elsavet Andrikopoulou, Kirsten Smith](#)

It is your responsibility to follow the University Code of Practice on Ethical Standards and any Department/School or professional guidelines in the conduct of your study including relevant guidelines regarding health and safety of researchers including the following:

- [University Policy](#)
- [Safety on Geological Fieldwork](#)

It is also your responsibility to follow University guidance on Data Protection Policy:

- [General guidance for all data protection issues](#)
- [University Data Protection Policy](#)

Which school/department do you belong to?: **School of Computing**

What is your primary role at the University?: **Postgraduate Student**

What is the name of the member of staff who is responsible for supervising your project?: **Dr. Alice Good**

Is the study likely to involve human subjects (observation) or participants?: No

Will financial inducements (other than reasonable expenses and compensation for time) be offered to participants?: No

Are there risks of significant damage to physical and/or ecological environmental features?: No

Are there risks of significant damage to features of historical or cultural heritage (e.g. impacts of study techniques, taking of samples)?: No

Does the project involve animals in any way?: No

Could the research outputs potentially be harmful to third parties?: No

Could your research/artefact be adapted and be misused?: No

Will your project or project deliverables be relevant to defence, the military, police or other security organisations and/or in addition, could it be used by others to threaten UK security?: No

Please read and confirm that you agree with the following statements: I confirm that I have considered the implications for data collection and use, taking into consideration legal requirements (UK GDPR, Data Protection Act 2018 etc.), I confirm that I have considered the impact of this work and and taken any reasonable action to mitigate potential misuse of the project outputs, I confirm that I will act ethically and honestly throughout this project

Supervisor Review

As supervisor, I will ensure that this work will be conducted in an ethical manner in line with the University Ethics Policy.

Supervisor comments:

Supervisor's Digital Signature: **alice.good@port.ac.uk**

Date: **15/09/2025**



School of Computing Postgraduate Programme

MSc in Data Analytics

Project Specification

Bovolenta, Adamo – UP2069775

Project Specification

Basic details

Student name:	Adamo Bovolenta
Draft project title:	AI-Enhanced Damage Assessment from Satellite Imagery in the Russia–Ukraine Conflict
Course and year:	MSc Data Analytics, 2024/25
Client organisation:	N/A
Client contact name:	N/A
Project supervisor:	Dr. Good, Alice

Outline of the project environment

Who is the client? What do they do? What is their problem? Why does it need to be solved?

Given the nature of this research as an academic dissertation, there is no designated external client. However, the intended beneficiaries of this study include humanitarian organisations, intervention teams, NGOs, and governmental agencies involved in response and reconstruction efforts in regions of the country affected by conflict.

The problem to be solved

What are the aims and objectives of the project?

What constraints are there on your solution to the problem?

The primary aim of this MSc research is to develop and assess a semantic segmentation model based on the U-Net architecture, aimed at accurately detecting and classifying infrastructure damage in satellite imagery of Donetsk Oblast. The model will be trained on annotated Sentinel-2 data, and its performance will be evaluated using conventional metrics such as Intersection over Union (IoU), precision, recall, and F1-score. The project's deliverables will include georeferenced damage-assessment maps, each delineating buildings classified in four damage categories: none, minor, major or destroyed.

These maps will incorporate intuitive colour schemes and legends to facilitate interpretation by users with varying levels of technical expertise, thereby enabling more efficient prioritisation of relief and reconstruction activities. The ideal outcome will be a comprehensive Master's thesis that documents the research thoroughly, starting from a review of the literature revolving

around semantic segmentation techniques and existing damage-assessment frameworks in conflict zones, continuing with a dissection of the methods used to deploy the model in discussion in this research, and present the results by quantitative metrics and qualitative map example, with a conclusive discussion of the results obtained.

Several practical and temporal constraints frame the scope of this research. The foremost challenge pertains to the availability and quality of satellite data. Although Sentinel-2 imagery is freely accessible, its 10-meter spatial resolution limits the discernment of fine structural details, and persistent cloud cover can reduce data usability.

Additionally, creating reliable ground-truth annotations is resource-intensive, requiring manual labelling of numerous buildings with expert judgment. Although existing datasets such as xBD and UNOSAT polygons can partially alleviate this burden, they do not fully eliminate the need for manual input.

Computational limitations also pose significant hurdles. The training and fine-tuning of deep-learning models demand substantial GPU resources and storage, which restricts hyperparameter tuning, the complexity of model ensembles, or the adoption of larger architectures for marginally better accuracy.

The project timeline emphasises the importance of experimentation and producing a publishable research article, rather than developing a fully polished, production-ready software solution. Nevertheless, a dashboard for visualising damage maps may be incorporated if time permits.

Lastly, field validation within the conflict zone is infeasible due to safety and access issues; consequently, evaluation will depend solely on secondary data sources and high-resolution imagery archives.

Breakdown of tasks

What is going to be your approach? What background research do you need to do? What tools do you need to acquire? What skills do you require, and how are you going to acquire those that you do not already have? What do you need to design and build?

The project will follow a structured workflow that begins with the collection of pre- and post-conflict satellite imagery specific to Donetsk Oblast and culminates in the creation of georeferenced damage-assessment maps, which will be detailed in the subsequent research paper.

Initially, the project will focus on reviewing key literature related to prominent semantic segmentation architectures, including variations of U-Net, DeepLab, and Siamese CNN. This review will be complemented by a comprehensive analysis of established frameworks for remote-sensing damage assessment, such as the xView2/xBD and UNOSAT datasets, to determine the most suitable methodologies for achieving the project's objectives.

Additionally, the review will examine how damage assessment techniques have evolved over time, particularly in the context of past conflicts between 2000 and 2024. Important cases such

as the Syrian Civil War, the Nagorno-Karabakh conflict, and the Gaza conflict will be examined to identify trends and improvements in remote sensing technology, data quality, and analysis models.

The project will also address best practices in geospatial preprocessing, such as cloud masking, orthorectification, and image co-registration, alongside annotation techniques to ensure accurate and reproducible results.

Key skills required for this project include advanced Python programming, geospatial data analysis, and designing deep-learning models. The project aims to enhance knowledge in semantic segmentation by following TensorFlow and PyTorch tutorials, replicating basic U-Net models, and gaining hands-on experience with remote-sensing data. For geospatial preprocessing, practical guides on tools such as GDAL and Rasterio will be utilised, with data visualisations performed using QGIS.

The workflow will involve scripts for downloading Sentinel-2 imagery, preprocessing steps like reprojection, cloud masking, and alignment, as well as an annotation pipeline combining existing building footprint shapefiles with manually created masks. Additionally, the code for training and evaluating the model will be implemented to produce quantitative metrics and qualitative segmentation maps. Post-processing routines will convert pixel-level predictions into classifications of building damage.

A Jupyter Notebook repository will offer interactive figures and maps for the thesis and serve as a reference for future researchers or humanitarian practitioners to build upon. The main tools supporting this process will include the Google Earth Engine Python API, GDAL, Rasterio, GeoPandas, Shapely, and deep learning frameworks such as TensorFlow and PyTorch. Development will be carried out using Visual Studio Code, Anaconda, and Jupyter Notebook, on a MacBook Air M2. The satellite data will be sourced chiefly from Sentinel-1 and Sentinel-2 satellites under the Copernicus program.

Project deliverables

What information system artefacts will be developed/adapted? What documents will be produced?

The research will produce an adaptable and efficient processing pipeline built with Jupyter Notebooks and supporting Python modules.

This pipeline will feature a data-ingestion component that connects to the Google Earth Engine API to retrieve and store Sentinel-2 images. It will also include a geospatial preprocessing suite employing GDAL, Rasterio, and GeoPandas to handle tasks such as projection alignment, cloud masking, and image registration.

Furthermore, a semantic segmentation model based on U-Net, implemented in both TensorFlow and PyTorch, will serve as the core analysis engine. Utility functions for training will include hyperparameter management, checkpoint saving, and metrics calculation. After training, the system will convert pixel-level damage predictions into vector outlines of buildings.

To ensure that these computational tools are easy to reproduce and use in different

environments, they will be packaged with an environment specification, using a requirements.txt file or a Dockerfile. If resources allow, a simple web interface built with Dash, Streamlit, or Tableau may be developed to enable non-technical users to view and filter damage maps interactively. Supporting documentation will detail the software system, including installation guides, data handling procedures, and model operation instructions. The main thesis report will include a comprehensive literature review, methodology, experimental results with performance metrics, and visual examples of damage maps.

Two detailed documentation volumes will also be produced: one for users describing how to operate the system, and another focused on reproducing the results, outlining environment setup and workflow steps.

Requirements

What are the client's requirements? or

How are you going to elicit the client's requirements?

Although there is no direct client involved, the project aims to support humanitarian and reconstruction agencies operating in Donetsk Oblast. These agencies require a fast, accurate method to assess and classify infrastructure damage into four categories: none, minor, major, and destroyed. The solution will use open-access satellite imagery to process, analyse, and generate visual damage maps and vector files compatible with typical Geographic Information System (GIS) platforms.

The system must adhere to key operational requirements, including being open-source and easily reproducible, particularly in environments with limited resources. Deployment should be simplified through containerization (e.g., Docker), allowing agencies to rerun analyses as new data becomes available without extensive additional work. These needs align with the standards outlined by humanitarian mapping guidelines, such as those from UN OCHA, and are designed to ensure that the system's capabilities meet operational demands of timeliness, clarity, data interoperability, and reproducibility, all while avoiding the necessity for direct on-field validation.

Legal, ethical, professional, social issues

What are the legal/ethical/professional/social issues that may impose constraints on the project?

How will you ensure that they will be complied with, or what steps will you take to avoid/mitigate their effects?

What research ethics approval (if any) is needed for your work and what will you do to obtain it?

The project must navigate various legal, ethical, professional, and social factors that could influence its implementation. Ensuring compliance and mitigating potential issues will involve following established guidelines and best practices throughout the research process.

Legal and Licensing

All satellite imagery and annotated datasets will be used strictly according to their licensing agreements. Sentinel-2 and Sentinel-1 data, accessible via the Copernicus Open Access Hub or Google Earth Engine, are available under open licenses that may necessitate appropriate attribution. The xBD dataset is distributed under the CC BY-NC-SA 4.0 license, permitting non-commercial academic use, with the requirement to share any derivative works under the same license. To adhere to these terms, all scripts and data derivatives will include explicit attribution statements and licensing information. Redistribution of data will also comply with original licensing conditions.

Ethical and “Don’t Harm” Principles

Automated damage assessment outputs could potentially be misused—for example, for military targeting or to stigmatise specific areas. While this project necessarily involves **individual building-level classification** to support precise humanitarian response and reconstruction planning, careful steps will be taken to reduce ethical risks. Model confidence scores and uncertainty estimates will accompany each classification to help users interpret predictions with appropriate caution. All visual materials will be anonymised and will **exclude sensitive contextual information**, such as facility names or coordinates related to shelters or strategic infrastructure. The purpose of this work is strictly humanitarian, and its outputs are designed to support transparent, responsible use by relief agencies and policymakers.

Professional and Reproducibility Standards

The project will follow rigorous open science practices. This includes using version control with Git, clearly specifying computational environments via requirements files or Docker containers, and documenting data provenance and preprocessing workflows thoroughly. These measures will facilitate reproducibility, enable peer review, allow bias auditing, and support future extensions or adaptations in other regions.

Social Responsibility

Given that the damage assessments are intended to inform humanitarian aid efforts in Donetsk Oblast, the results must be accurate and responsibly used. Inaccurate or misunderstood outputs could lead to inefficient use of limited resources. To encourage proper usage, clear guidelines will be provided for interpreting and applying the maps, along with contextual information such as the date of image capture, cloud cover conditions, and the model’s performance metrics.

Research Ethics Approval

Since the research relies solely on publicly accessible, non-identifiable satellite images and secondary annotated datasets—without involving human subjects or personal data—formal Institutional Review Board (IRB) approval is not required.

Facilities and resources

What computing/IT facilities will you use/require?

What other facilities/resources will you use/require?

Are there constraints on their availability? If funds are required to acquire them, have these been allocated? Will they be available in time?

The project will utilise a combination of local and remote computing resources to support its computational needs.

Primary Computing Facilities

The main development work will be conducted on a local workstation running macOS or a university-provided desktop running Windows/Linux OS. Essential software includes Visual Studio Code or PyCharm for coding, Jupyter Notebook or Lab for interactive analysis, and Docker for containerised reproducibility. The hardware specifications should include at least an eight-core CPU, 32 GB of RAM, a high-performance NVIDIA GPU with 10 GB or more VRAM (critical for training deep learning models), and a minimum of 1 TB SSD storage (either on a local drive or external solid-state device) to accommodate raw Sentinel-2 data and intermediate analysis files.

Additional Computing Facilities

For large-scale data processing or when extra GPU capacity is required, the project will leverage the University of Portsmouth's HPC cluster via SSH or JupyterHub, provided the servers are operational. This facility offers multi-node distributed computing for tools like Hadoop and PySpark, as well as additional GPU nodes. Internet access through wired connection or eduroam and VPN connectivity, including necessary hostnames (e.g., mn01.soc.uni.ds.port.ac.uk), have already been established and incur no extra costs.

Software and Licensing

All necessary libraries—such as GDAL, Rasterio, GeoPandas, Shapely, NumPy, Pandas, OpenCV, Matplotlib, Seaborn, TensorFlow, Keras, PyTorch, Scikit-learn, and Albumentations—are open-source and can be installed via pip or conda. Annotation tools like QGIS and CVAT, as well as the Google Earth Engine Python API, are also freely available. Docker support is ensured on both local systems and HPC cluster environments. No additional funding for software licenses is required.

Potential Constraints and Mitigation Strategies

The main potential bottleneck is GPU availability on the HPC cluster during peak hours. To address this, scheduling training jobs during off-peak times or using cloud alternatives like Google Colab Pro (with existing departmental credits) will ensure continued progress. All hardware and software resources are either already allocated or available at no cost.

Cloud-Based Alternatives

Should the university's Linux cluster become permanently unavailable, the project will shift entirely to cloud-based or local resources, without disrupting timelines or incurring extra costs.

Prototype development, exploratory training, and hyperparameter tuning can be performed on Google Colab Pro or Colab Pro+, which provide free access to NVIDIA GPUs and TPUs.

For time-consuming training tasks exceeding Colab's session limits, the pipeline can be deployed on managed GPU instances from AWS, GCP, or Microsoft Azure. These platforms offer spot instances and on-demand VMs equipped with NVIDIA hardware. Data can be stored and managed via cloud object storage services like S3 or GCS, replacing traditional HDFS setups. Docker containers will be used to replicate consistent development environments, with research credits covering these cloud services, eliminating additional expenses.

Project plan

What are you going to do when? (This may be an attached output from MS Project etc.)

What risks to the success of the project have you identified? What steps can you take to minimise them? What backup plans do you have if things identified go wrong?

Week 1 (June 09 - 15)

Conducting a targeted literature review focused on damage assessment techniques in the Russia–Ukraine conflict, with Donetsk Oblast as the main case study.

To understand how damage assessment practices and technologies have evolved in recent years, a comparative analysis of similar methods used in other major conflicts between 2000 and 2024, such as the Syrian Civil War, Gaza conflicts, and Iraq-based operations, will be performed.

Examination of the **technological progression** in:

- **Satellite image resolution** (e.g. Landsat vs. Sentinel vs. commercial providers)
- **Deep learning models** for semantic segmentation (U-Net, DeepLabV3, Siamese CNN)
- **Frameworks and pipelines** used for remote sensing damage analysis

Reviewing damage assessment datasets: xBD, UNOSAT, xView2

Reviewing segmentation architectures: U-Net, DeepLab, Siamese CNN

Assessing Sentinel-2 availability for Donetsk (cloud cover, seasonality)

Defining criteria for evaluating model output and visualisation quality:

- Quantitative metrics (IoU, F1, recall/precision)
- Qualitative expectations for **map visualisation clarity and interpretability**

Supervisor Meeting #1

Week 2 (June 16–22)

Environment Setup: Configuring Python environment with key geospatial and ML libraries (GDAL, Rasterio, Earth Engine API, TensorFlow/PyTorch) using Conda or Docker. Set up VS Code, Jupyter, and Git.

GEE Workflow: Using Google Earth Engine Python API to select, filter (by date, cloud cover), and download **pre/post-conflict Sentinel-2 imagery** for Donetsk Oblast. Focusing on regions with active conflict and clear damage visibility.

Image Pairing & Organisation: Script a pipeline to fetch, log, and store **temporally matched image pairs**. Automating metadata capture (e.g., dates, cloud score) for traceability.

Initial Preprocessing:

Using Sentinel's QA60 band or Scene Classification Layer (SCL) to apply cloud masking.

Performing **coregistration** to spatially align image pairs via affine transformations and Rasterio/GDAL.

Validating image alignment manually (QGIS overlays) and statistically (histogram comparison).

Output: Exporting 3–5 co-registered image pairs and visualising in QGIS to confirm alignment and data quality.

Code Management: Begin modularising scripts, with version control on GitHub for transparency and reproducibility.

Week 3 (June 23–29) – Annotation Pipeline & Preprocessing:

Label Integration: Importing and aligning **xBD damage annotations** with pre/post-event Sentinel-2 imagery. Matching geospatial references, projections, and scales.

Mask Generation: Converting vector labels (e.g., building polygons) into **pixel-wise segmentation masks**, encoding damage classes (none, minor, major, destroyed). Ensuring masks align precisely with imagery grids.

Pipeline Refinement:

- Updating and modularising preprocessing scripts to automate:
 - AOI clipping
 - Cloud masking
 - Projection alignment
 - Rasterisation of labels

Visual Validation: Loading co-registered images and generated masks into **QGIS** to visually inspect alignment and labelling quality.

Output: Producing 5–10 well-aligned annotated image/mask pairs for training/test set prototyping.

Milestone: Conducting **supervisor meeting #2** to review preprocessing pipeline, data quality, and annotation strategy.

Week 4 (June 30–July 6) – Model Prototyping (U-Net):

Model Implementation: Building a **baseline U-Net architecture** in TensorFlow or PyTorch for semantic segmentation of building damage.

Initial Training: Training the model on a **small, annotated subset** (200–300 buildings) using paired images and damage masks.

Monitoring & Evaluation: Tracking key training metrics — **loss, accuracy, and confusion matrix** — using callbacks or logging tools.

Code Modularity: Developing reusable modules for:

- Training and validation loops
- Metric computation (IoU, precision, recall)
- Model checkpointing and loading

Early Outputs: Saving initial segmentation results (images + masks) and checkpoints to evaluate model performance visually and quantitatively.

Week 5 (July 7–13) – Model Evaluation + Refinement

Validation Testing: Evaluating model performance on a **held-out validation set**, using pixel-wise accuracy, IoU, and damage-class precision/recall.

Hyperparameter Tuning: Experimenting with learning rate, batch size, and augmentation strategies to improve generalisation.

Framework Comparison (Optional): Benchmark **TensorFlow** vs. **PyTorch** implementations for speed, accuracy, and ease of integration.

Damage Aggregation Logic: Implementing mapping from **pixel-level outputs to building-level classifications** using vector overlays and majority class rules.

Supervisor Meeting #3: Presenting model results, tuning strategy, and classification logic for feedback and validation.

Week 6 (July 14–20) – Visualisation + Thesis Writing 1

Map Generation: Producing **georeferenced output maps** with damage-level overlays using segmentation results and building footprints.

Demo Dashboard: Building a **basic interactive dashboard** (Streamlit/QGIS plugin/Tableau) to visualise and explore damage maps.

Thesis Writing – Methodology Chapter:

Data sourcing (Sentinel-2, UNOSAT, xBD)

Preprocessing steps (cloud masking, co-registration, clipping)

Model architecture (U-Net details)

Training workflow (loss functions, metrics, augmentation)

Version Control: Pushing all scripts, notebooks, and processing modules to **GitHub** for reproducibility and backup.

Week 7 (July 21–27) – Thesis Writing II + Reproducibility

Results & Discussion Writing:

- Presenting evaluation **metrics** (IoU, precision, recall)
- Including **visual outputs** (segmentation maps, overlays)
- Critically assessing **model limitations** and sources of error

Reproducibility Tools:

- Finalising requirements.txt or Dockerfile for environment setup
- Ensuring all dependencies and paths are documented

Documentation:

- Writing a **user guide** for running the pipeline and interpreting outputs
- Adding technical documentation for preprocessing, model usage, and inference

Supervisor Meeting #4: Submitting full pipeline and draft thesis for review and feedback

Deliverables:

- **Full draft of the dissertation**, with figures, tables, and references
- Complete codebase + docs on GitHub

Week 8 (July 28–Aug 3) – Review & Buffer Week

Thesis Revisions: Applying the supervisor’s feedback to refine text, structure, and clarity across all sections.

Visual Enhancements: Improving the quality and consistency of **figures, maps, and tables** for a final presentation.

Reproducibility Check: Testing the entire codebase from scratch using **requirements.txt** or **Dockerfile** to ensure **full environment replicability**.

Submission Prep: Finalising all documents, visuals, and GitHub repository; organising them into a complete **submission-ready package** by early August.

Optional Enhancement: Refining and polishing the **interactive dashboard** (Streamlit/QGIS) for demo or appendix inclusion.

Supervision meetings

(Very important) What general plan has been agreed with your supervisor regarding supervision meetings over the period of the project? A detailed schedule of meetings is not needed here, but you should have agreed on the frequency of meetings and the mode of meeting (face-to-face where possible, video conferencing, telephone, email, etc). How will your project plan accommodate any extended absences of your supervisor (annual leave, etc)?

Supervision will occur fortnightly, utilising standard video conferencing and face-to-face meetings when necessary. Between scheduled meetings, progress updates and ad-hoc questions will be addressed via email to ensure continuity.

Project mode

If there are two possibilities for your project mode, after negotiation, please record your planned duration and submission date. It is also helpful to record your initial registration mode (i.e. are you a full time or a part time student). Remember, the exact dates will be announced through Moodle. These represent a generic guideline.

Registration mode

Full Time	
Full Time	
11/9/2025	

Project mode

Planned submission deadline

Signatures

Signature: _____

Date: _____

Student	Adamo Bovolenta	03/06/2025
Client	N/A	
Project supervisor	Alice Good	03/06/2025